



A

PROJECT REPORT ON

“CARVION AI – AI-POWERED CAREER DEVELOPMENT PLATFORM”

SUBMITTED FOR THE PARTIAL FULFILMENT OF THE

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SUBMITTED BY

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UNDER THE GUIDANCE OF

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M.P.E SOCIETY'S S.D.M. DEGREE COLLEGE AND PG STUDIES,

HONNAVAR

ACCREDITED WITH "A+" GRADE BY NAAC

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M.P.E Society's
S.D.M DEGREE COLLEGE

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Carvion AI – AI-Powered Career Development Platform

For the said Degree from S.D.M Degree College, Honnavar

In the year of March to June 2026

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LETTER OF CONFIRMATION

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during the academic year March to June 2026.

Date:

Place:

Signature of HOD

Signature of Principal



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DECLARATION

I hereby declare that the Project Report title “Carvion AI – AI-Powered Career Development Platform” has been prepared by Mr. Lohith Mableshwara Marathi during the year 2026 under valuable guidance and supervision of Prof. Vandana Hegde (HOD of MCA, S.D.M Degree College, Honnavar) for the fulfillment of the requirement of Masters of Computer Applications.

I also declare this Project is the result of my own effort and has not been submitted to any other university for the award of any degree.

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Finally, I take this opportunity to express our gratitude and respect to all those who are directly and indirectly helped and encouraged us during the course of project.

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ABSTRACT

Carvion AI is a full-stack, AI-powered career development platform developed to assist job seekers in improving their resumes, identifying suitable career paths, and preparing for job interviews through a unified web-based system. The platform combines modern web technologies with Artificial Intelligence (AI) to provide personalised career guidance, resume analysis, learning recommendations, and interview preparation.

The system follows a client-server architecture in which the frontend is developed using React.js with Vite, while the backend is implemented using Python, Django, and Django REST Framework. User authentication is managed using JSON Web Token (JWT). Application data is stored in MongoDB using the MongoEngine Object Document Mapper (ODM), whereas administrator information and session-related data are maintained in SQLite. AI-based features are implemented using the Google Gemini 2.5 Flash API, which enables intelligent analysis and personalised recommendations.

platform consists of six major modules. The Resume Intelligence module extracts information from uploaded resumes, calculates ATS compatibility scores, identifies missing skills, and provides suggestions for improvement. The Career Roadmap module generates personalised learning paths based on the user's career goals and existing skills. The Mock Interview module simulates interview sessions and provides AI-based feedback on candidate responses. The Job and Course Recommendation module suggests relevant job opportunities and learning resources according to the user's profile. The AI Chatbot Assistant helps users navigate the platform and answers career-related queries. The Admin Module enables user management, system monitoring, notification management, activity tracking, and dashboard-based reporting using real-time application data.

The system incorporates several security features, including role-based access control, JWT-based authentication, request validation, API rate limiting, and application logging to improve security and system reliability. In addition, if the AI service is temporarily unavailable, the platform uses a fallback mechanism to maintain the availability of essential features. Overall, Carvion AI provides an integrated solution for resume improvement, career planning, skill development, job recommendations, and interview preparation within a single intelligent platform.

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INTRODUCTION

The rapid growth of information technology and Artificial Intelligence (AI) has significantly transformed the recruitment process and career development practices. Today, job seekers face several challenges, such as preparing professional resumes, passing Applicant Tracking Systems (ATS), identifying the skills required for specific job roles, selecting suitable learning resources, and performing well in technical and HR interviews. Traditional career guidance methods and online job portals generally provide limited support, as they focus on individual services rather than offering a complete career development solution.

Carvion AI is developed to address these challenges by providing an integrated, AI-powered career development platform. The system combines resume analysis, career planning, skill gap identification, job recommendations, learning recommendations, mock interview preparation, and AI chatbot assistance within a single web application. By analysing the user's profile and career goals, the platform provides personalised recommendations to support continuous career growth.

The platform makes use of Artificial Intelligence through the Google Gemini 2.5 Flash API to analyse resumes, generate career roadmaps, recommend relevant skills and learning resources, evaluate interview responses, and answer career-related questions. These AI-based features help users receive personalised guidance and improve their job readiness. The system is designed to provide intelligent recommendations while maintaining ease of use and reliable performance.

From a technical perspective, Carvion AI follows a client-server architecture. The frontend is developed using React.js with Vite, providing a responsive and interactive user interface. The backend is implemented using Python, Django, and Django REST Framework, which exposes RESTful APIs for communication between the client and the server. MongoDB, using the MongoEngine Object Document Mapper (ODM), is used to store user profiles, resume information, AI-generated recommendations, and other application data. SQLite is used to maintain administrator information and session-related data. User authentication is implemented using JSON Web Token (JWT) authentication, while role-based access control ensures that standard users and administrators have appropriate access permissions.

The platform also includes an Admin Module that enables administrators to manage user accounts, monitor platform activities, send notifications, view system reports, and supervise the overall operation of the application through real-time data. This module helps maintain the security, reliability, and proper management of the platform.

This project demonstrates the practical application of modern web development technologies and Artificial Intelligence in solving real-world career development problems. The report presents the system requirements, software design, architecture, implementation, testing, and evaluation of Carvion AI, highlighting its role as an intelligent career guidance platform for students and job seekers.

MOTIVATION

The primary motivation for developing Carvion AI is to address the challenges faced by job seekers during the recruitment and career development process. In today's competitive job market, candidates are expected to prepare professional resumes, possess relevant technical and soft skills, and perform well in interviews. Many organizations use Applicant Tracking Systems (ATS) to screen resumes before they are reviewed by recruiters. As a result, even qualified candidates may not be shortlisted if their resumes do not meet ATS requirements. Most online resume builders only help create resumes but do not provide detailed analysis, ATS evaluation, or suggestions for improvement.

Another important challenge is the lack of an integrated career development platform. Job seekers often use different websites and applications for resume creation, job searching, online learning, interview preparation, and career guidance. This makes the career preparation process time-consuming and less efficient, as users must repeatedly provide their information and manage their progress across multiple platforms.

Professional career counselling and interview coaching can help candidates improve their job readiness; however, these services are often expensive and not easily accessible to students, fresh graduates, and early-career professionals. Therefore, there is a need for an affordable and intelligent system that provides personalised career guidance through a single platform.

Carvion AI is developed to address these limitations by integrating resume analysis, ATS score evaluation, skill gap identification, career roadmap generation, job recommendations, learning recommendations, AI-based mock interviews, and chatbot assistance into one web application. The platform uses the Google Gemini 2.5 Flash API to provide intelligent recommendations and personalised career guidance based on the user's profile and career goals. By combining these features, Carvion AI aims to simplify the career development process and help users improve their employability and interview readiness through a unified AI-powered platform.

OBJECTIVES

The main objectives of the Carvion AI project are as follows:

1. To Develop an AI-Based Resume Analysis System

The objective of this module is to develop a system that accepts resumes in different formats, extracts relevant information, and analyses the content using Artificial Intelligence. The system evaluates the resume for Applicant Tracking System (ATS) compatibility, identifies missing skills, and provides suggestions to improve the quality and effectiveness of the resume.

2. To Generate Personalised Career Roadmaps

The project aims to provide users with customised career development plans based on their existing skills, qualifications, and target job roles. The system recommends learning paths, important skills, certifications, and project ideas that help users achieve their career goals in a structured manner.

3. To Implement an AI-Based Mock Interview System

The objective is to create an intelligent interview preparation module that presents role-specific interview questions and evaluates the user's responses using Artificial Intelligence. The system provides constructive feedback on technical knowledge, communication skills, and areas that require improvement.

4. To Provide Intelligent Job and Course Recommendations

The platform is designed to recommend suitable job opportunities and online learning resources according to the user's skills, career interests, and learning requirements. This helps users improve their knowledge while simultaneously identifying employment opportunities that match their profiles.

5. To Develop an AI Chatbot for Career Guidance

The objective is to integrate an AI-powered chatbot that provides instant assistance to users by answering career-related questions, explaining resume analysis results, suggesting learning resources, and guiding users through different features of the platform.

6. To Implement Secure User Authentication and Authorization

The project aims to develop a secure authentication system using JSON Web Tokens (JWT) and role-based access control. This ensures that user information is protected and that administrators and standard users have appropriate access to system functionalities.

7. To Develop an Efficient Administration Module

The objective is to provide administrators with a dedicated module for managing users, monitoring platform activities, viewing reports, sending notifications, and supervising the overall operation of the system. This improves the management, maintenance, and reliability of the application.

8. To Develop a Full-Stack Web Application

The project aims to implement a modern client-server architecture using React.js with Vite for the frontend and Python, Django, and Django REST Framework for the backend. The frontend and backend communicate through RESTful APIs to ensure efficient data exchange and smooth application performance.

9. To Ensure System Security, Reliability, and Performance

The objective is to enhance the security and reliability of the platform by implementing request validation, API rate limiting, application logging, and fallback mechanisms for AI services. These features help protect user data and maintain uninterrupted system operation.

10. To Improve Career Development Through Artificial Intelligence

The final objective is to provide students and job seekers with an integrated platform that combines resume analysis, career planning, skill development, job recommendations, interview preparation, and AI assistance. This enables users to prepare more effectively for employment opportunities using a single web-based application.

LITERATURE SURVEY

The development of AI-based career guidance systems combines several research areas, including resume information extraction, Applicant Tracking System (ATS) matching, career recommendation, and conversational Artificial Intelligence. Recent studies show that AI techniques can improve the accuracy of resume analysis, personalise career recommendations, and provide better support for interview preparation and career planning.

1. Resume Information Extraction

Resume parsing converts unstructured documents such as PDF and DOCX files into structured information. Earlier approaches relied on keyword matching and regular expressions, which were less accurate for different resume formats. Recent research uses Artificial Intelligence and Natural Language Processing (NLP) techniques, including Transformer-based models, to extract information such as personal details, education, work experience, and technical skills more accurately.

2. ATS Compatibility and Resume Matching

Modern Applicant Tracking Systems (ATS) use semantic similarity techniques instead of simple keyword matching. Transformer-based language models compare resumes with job descriptions by understanding the meaning of the text rather than matching exact words. This approach improves resume evaluation and provides more accurate ATS scores and skill gap analysis.

3. Career Path Recommendation

Research in career recommendation focuses on identifying skill gaps and suggesting suitable learning paths based on a user's current skills and career goals. AI-based recommendation systems analyse user profiles and recommend courses, certifications, and career roadmaps to help users achieve their desired job roles.

4. AI-Based Mock Interviews

Artificial Intelligence has also been applied to interview preparation systems. AI-powered interview platforms generate role-specific interview questions, evaluate user responses, and provide feedback on technical knowledge and communication skills. These systems help users improve their interview performance through personalised practice sessions.

OUTCOME OF LITERATURE SURVEY

The literature survey provided valuable insights into the design and development of AI-based career guidance systems. The major outcomes obtained from the study are as follows:

1. Improved Resume Analysis

Recent Artificial Intelligence and Natural Language Processing (NLP) techniques provide more accurate resume parsing and information extraction than traditional keyword-based methods.

2. Semantic ATS Evaluation

Modern ATS systems use semantic analysis instead of simple keyword matching, resulting in more accurate resume evaluation, skill gap identification, and personalised recommendations.

3. Need for an Integrated Platform

Existing systems mainly focus on individual features such as resume analysis, career guidance, or interview preparation. The literature highlighted the need for a single platform that combines these functionalities.

4. Importance of Secure and Scalable Design

Research also emphasised the need for secure authentication, efficient database management, administrative control, and scalable system architecture for reliable platform operation.

Based on these findings, the Carvion AI Career Development Platform was designed to integrate resume analysis, career roadmap generation, mock interview preparation, job recommendations, learning recommendations, and AI chatbot assistance into a single web-based application. The system uses the Google Gemini 2.5 Flash API to provide intelligent career guidance while maintaining a secure, efficient, and user-friendly environment.

EXISTING SYSTEM

The existing career development system consists of different online platforms that provide individual services such as resume creation, job searching, online learning, and interview preparation. Although these platforms are useful, they work independently and do not provide a complete career guidance solution. The major components of the existing system are explained below.

1. Resume Building Platforms

Resume builders such as Canva, Zety, and Novoresume help users create professional resumes using ready-made templates. These platforms mainly focus on resume formatting and design. However, they do not evaluate ATS compatibility, analyse resume content, identify missing skills, or provide personalised recommendations for improvement.

2. Job Portal Platforms

Job portals such as LinkedIn, Indeed, and Naukri allow users to search and apply for jobs in different industries. They provide access to numerous job opportunities but do not evaluate whether a candidate is suitable for a particular role. They also do not suggest the skills required to improve job selection chances.

3. Online Learning Platforms

Learning platforms such as Coursera, Udemy, and edX offer various online courses and certification programs. These platforms help users improve their knowledge but do not automatically recommend learning paths based on a user's existing skills or career objectives. Users must decide independently which courses are suitable for them.

4. Interview Preparation Platforms

Platforms such as Pramp and Interview Cake provide interview practice through coding challenges, mock interviews, and sample interview questions. Most of these systems focus mainly on technical interview preparation and do not provide AI-based evaluation or personalised feedback on user responses.

5. Career Guidance Platforms

Some websites provide career advice, resume tips, and interview guidance through blogs or static content. These platforms provide general information but do not offer personalised career planning, skill gap analysis, or AI-assisted recommendations based on individual user profiles.

LIMITATIONS OF THE EXISTING SYSTEM

1. Lack of Integration

Existing systems perform only one specific function, such as resume building, job searching, learning, or interview preparation. Users must use multiple platforms to complete their career development process, making the workflow complex and time-consuming.

2. Limited Resume Analysis

Most resume builders only create resumes and provide formatting options. They do not analyse resume quality, evaluate ATS scores, detect missing skills, or provide intelligent suggestions for improvement.

3. Limited Personalisation

The recommendations provided by existing systems are mostly general and are not customised according to the user's education, skills, work experience, or career goals. This reduces the effectiveness of career guidance.

4. No Complete Skill Gap Analysis

Most existing platforms do not compare the user's current skills with the requirements of the target job role. Therefore, users cannot easily identify which technical or professional skills need improvement.

5. Lack of AI-Based Career Guidance

Many traditional career platforms do not use Artificial Intelligence to provide personalised career recommendations, resume evaluation, interview feedback, or learning suggestions.

6. Separate Learning and Job Search

Learning platforms and job portals work independently. There is no direct connection between learning new skills and recommending suitable job opportunities based on those newly acquired skills.

7. Limited Interview Evaluation

Most interview preparation platforms provide only practice questions or sample answers. They do not evaluate user responses using Artificial Intelligence or provide detailed feedback on communication skills, technical knowledge, and confidence.

PROPOSED SYSTEM

The proposed system, Carvion AI, is an AI-powered career development platform that integrates resume analysis, career guidance, skill development, interview preparation, and job recommendations into a single web application. The platform uses Artificial Intelligence to provide personalised career guidance and helps users improve their employability through an integrated and user-friendly system.

The major features of the proposed system are described below.

1. Unified Career Development Platform

The proposed system combines multiple career development services within a single application. Users can upload resumes, analyse their skills, generate career roadmaps, prepare for interviews, search for jobs, explore learning resources, and communicate with an AI chatbot without using multiple platforms. A single user profile is used across all modules, providing a consistent and personalised experience.

2. AI-Based Resume Analysis

The platform allows users to upload resumes in PDF and DOCX formats. Resume information is extracted using PyMuPDF and python-docx libraries. The extracted data is analysed using the Google Gemini 2.5 Flash API to identify skills, education, work experience, projects, and certifications. The system also calculates an ATS compatibility score, detects missing skills, and provides recommendations to improve the resume.

3. Personalised Career Roadmap Generation

Based on the user's current skills and target career role, the system generates a personalised career roadmap. The roadmap includes recommended technical skills, learning objectives, certifications, project suggestions, and an estimated learning sequence. This helps users plan their career development in a structured and organised manner.

4. AI-Based Mock Interview System

The proposed system provides an AI-powered mock interview module where users can practice interview questions related to their chosen job role. The AI evaluates the user's responses and provides feedback on technical knowledge, communication skills, confidence, and areas that require improvement. This enables users to improve their interview performance before attending real interviews.

5. Job and Learning Recommendation System

The platform recommends suitable job opportunities and online learning resources

based on the user's profile, skills, career goals, and resume analysis. These recommendations help users improve their qualifications while identifying employment opportunities that match their abilities.

6. AI Chatbot Assistant

An AI-powered chatbot is integrated into the platform to provide career guidance and answer user queries. The chatbot assists users by explaining resume analysis reports, suggesting learning resources, answering career-related questions, and helping users navigate different features of the application.

7. Administration Module

The system includes a dedicated Admin Module for administrators to manage users and monitor the overall platform. The module provides facilities for user management, dashboard reporting, notification management, activity monitoring, and system supervision using real-time application data.

8. Secure Authentication and Authorization

The application implements JSON Web Token (JWT) authentication and role-based access control to ensure secure user login and data protection. Different access permissions are provided for administrators and standard users, improving system security and protecting sensitive information.

9. Reliable and Secure System

The proposed system incorporates request validation, API rate limiting, application logging, and AI fallback mechanisms to improve system reliability. If the AI service is temporarily unavailable, the platform continues to provide essential services, ensuring uninterrupted system operation.

10. Modern Full-Stack Architecture

The system follows a client-server architecture using React.js with Vite for the frontend and Python, Django, and Django REST Framework for the backend. MongoDB with MongoEngine ODM is used for application data storage, while SQLite is used for administrator and session-related information. RESTful APIs enable secure communication between the frontend and backend.

ADVANTAGES OF THE PROPOSED SYSTEM

1. Integrated Career Development Platform

The platform combines resume analysis, career planning, interview preparation, job recommendations, learning resources, and AI assistance within a single application. This eliminates the need to use multiple websites or software.

2. AI-Based Personalised Guidance

Artificial Intelligence analyses the user's profile and provides personalised recommendations for resume improvement, career planning, interview preparation, and skill development.

3. ATS Resume Evaluation

The system evaluates resumes using ATS-based analysis and identifies missing skills, helping users improve their chances of being shortlisted by recruiters.

4. Personalised Career Roadmaps

The platform generates customised learning plans based on the user's existing skills and career objectives, enabling systematic career growth.

5. AI-Based Interview Preparation

Users can practice interviews and receive AI-generated feedback on their technical knowledge, communication skills, and overall interview performance.

6. Intelligent Job and Course Recommendations

The system recommends relevant jobs and learning resources according to the user's skills, interests, and career goals, improving the effectiveness of career preparation.

7. Secure User Management

JWT authentication and role-based access control ensure secure access to the application while protecting user information and maintaining data privacy.

8. Efficient Administration

The Admin Module enables administrators to manage users, monitor platform activities, send notifications, and generate reports, making system administration simple and efficient.

FEASIBILITY STUDY

A feasibility study was conducted to determine whether the proposed **Carvion AI** system can be successfully developed, implemented, and maintained. The study evaluates the project from three important perspectives: Technical Feasibility, Operational Feasibility, and Economic Feasibility.

TECHNICAL FEASIBILITY

Technical feasibility determines whether the required hardware, software, technologies, and technical resources are available for successful implementation of the system.

1. Modern Technology Stack

The system is developed using modern and reliable technologies such as React.js with Vite for the frontend and Python, Django, and Django REST Framework for the backend. These technologies are widely used for developing secure and scalable web applications.

2. Database Support

The application uses MongoDB with MongoEngine ODM to store user profiles, resumes, AI-generated recommendations, interview results, and other application data. SQLite is used for administrator information, session management, and authentication-related data.

3. Artificial Intelligence Integration

The platform integrates the Google Gemini 2.5 Flash API to perform resume analysis, ATS score calculation, career roadmap generation, interview evaluation, chatbot assistance, and recommendation services. This enables intelligent and personalised career guidance.

4. Document Processing

The system uses PyMuPDF for PDF processing and python-docx for DOCX document processing. These libraries efficiently extract resume information for further analysis.

5. Security and Authentication

The application implements JSON Web Token (JWT) authentication and role-based access control to provide secure user login, protect sensitive information, and ensure appropriate access permissions for users and administrators.

OPERATIONAL FEASIBILITY

Operational feasibility evaluates whether the proposed system can be effectively used, managed, and maintained by its intended users and administrators.

1. Easy-to-Use Interface

The platform provides a simple, responsive, and user-friendly interface that allows users to upload resumes, analyse their skills, explore career roadmaps, prepare for interviews, and receive AI-generated recommendations with minimal effort.

2. Integrated Career Development

Unlike existing systems that provide separate services, Carvion AI combines resume analysis, ATS evaluation, skill gap analysis, career planning, learning recommendations, job recommendations, mock interviews, and chatbot assistance into a single platform.

3. Administration Module

The system includes an Admin Module that allows administrators to manage user accounts, monitor platform activities, view reports, send notifications, and supervise the overall operation of the application using real-time data.

4. Reliable System Operation

The platform includes request validation, API rate limiting, application logging, and AI fallback mechanisms. If the AI service is temporarily unavailable, the system continues to provide basic functionality without interrupting the user experience.

5. Reduced Manual Work

Most career guidance activities, including resume evaluation, interview assessment, career roadmap generation, and recommendation services, are automated using Artificial Intelligence, thereby reducing manual effort and improving efficiency.

ECONOMIC FEASIBILITY

Economic feasibility determines whether the project can be developed and maintained within a reasonable cost while providing significant benefits to users.

1. Use of Open-Source Technologies

The project is developed using open-source technologies such as React.js, Vite, Python, Django, Django REST Framework, MongoDB, and SQLite. This eliminates software licensing costs and reduces the overall development cost.

2. Cost-Effective AI Integration

The system uses the Google Gemini 2.5 Flash API, which follows a pay-as-you-use pricing model. This allows AI features to be implemented without investing in expensive hardware or maintaining large AI models locally.

3. Low Infrastructure Cost

The application can run on standard computer systems and can be deployed using cloud services with moderate infrastructure requirements. This reduces hardware and maintenance expenses.

4. Reduced Human Effort

The automation of resume analysis, career guidance, interview evaluation, and recommendation services reduces the need for continuous manual career counselling, saving both time and operational costs.

5. Scalable Solution

The platform can support an increasing number of users without requiring significant additional resources. This makes the system suitable for educational institutions, training centres, and career guidance organizations.

METHODOLOGY

The development of Carvion AI follows the Agile Software Development Life Cycle (SDLC), which supports iterative development, continuous testing, and regular improvements throughout the project. The system was developed in multiple phases, allowing each module to be designed, implemented, tested, and refined before moving to the next stage. The methodology adopted for the development of the proposed system is described below.

1. Requirements Gathering and Analysis

The first phase involved collecting and analysing the requirements of the proposed system. Functional requirements such as user registration and login, resume upload, resume analysis, ATS score calculation, career roadmap generation, job recommendations, learning recommendations, AI chatbot, mock interviews, and administrator management were identified. Non-functional requirements such as security, system performance, reliability, usability, response time, and database consistency were also defined. This phase helped establish the overall scope and objectives of the project.

2. System Design and Architecture

After analysing the requirements, the overall architecture of the system was designed. The project follows a client-server architecture, where the frontend communicates with the backend through RESTful APIs. Data Flow Diagrams (DFDs) and Entity Relationship Diagrams (ERDs) were prepared to represent data flow and database structure. The database schema was designed using MongoDB for application data and SQLite for administrator and session-related information.

3. Database and API Development

In this phase, the backend of the application was developed using Python, Django, and Django REST Framework. APIs were created for user authentication, resume management, ATS analysis, career roadmap generation, job recommendations, chatbot interaction, and administrative operations. MongoEngine ODM was used to manage MongoDB collections, while JWT authentication was implemented to provide secure access to the application.

4. AI and Document Processing Integration

Artificial Intelligence features were integrated using the Google Gemini 2.5 Flash API. The system processes uploaded resumes in PDF and DOCX formats using PyMuPDF and python-docx. The extracted information is analysed to generate ATS scores, identify missing skills, recommend career roadmaps, evaluate interview responses, and provide chatbot assistance. This phase enabled intelligent and personalised career guidance throughout the platform.

5. Frontend Development

The user interface was developed using React.js with Vite, along with HTML5, CSS3, and JavaScript. The frontend communicates with the backend through Axios and consumes RESTful APIs for data exchange. The interface was designed to provide easy navigation, responsive layouts, and a user-friendly experience for accessing all career development features.

6. Testing and Quality Assurance

Various testing techniques were performed to ensure the correctness and reliability of the application. Unit testing verified individual modules, integration testing validated communication between frontend and backend components, system testing checked complete application workflows, security testing verified authentication and authorization mechanisms, and performance testing ensured acceptable response times and stable system performance.

7. Deployment and Maintenance

After successful testing, the application was prepared for deployment. Application logging, request validation, API rate limiting, and backup mechanisms were implemented to improve system reliability. The Admin Module provides facilities for user management, activity monitoring, notification management, and overall system supervision. Regular maintenance and future updates can be performed to enhance system performance and introduce new features.

REQUIREMENTS

The requirements of the proposed system are classified into Functional Requirements and Non-Functional Requirements.

FUNCTIONAL REQUIREMENTS

Functional requirements describe the services and operations that the system must perform to meet user and administrator needs.

1. User Authentication and Account Management

The system provides secure user authentication and profile management facilities.

Requirements:

- ❖ User registration using a valid email address and password.
- ❖ Secure login using JSON Web Token (JWT) authentication.
- ❖ Password change and profile update facilities.
- ❖ Role-based access control for users and administrators.
- ❖ Secure logout and session management.
- ❖ Restricted access to protected pages and APIs.

2. Resume Management

The system enables users to upload, analyse, and manage their resumes.

Requirements:

- ❖ Upload resumes in PDF and DOCX formats.
- ❖ Extract personal details, education, work experience, technical skills, projects, and certifications.
- ❖ Generate ATS compatibility scores.
- ❖ Identify missing skills and provide resume improvement suggestions.
- ❖ Store resume information securely in the database.
- ❖ View, update, download, and delete uploaded resumes.

3. Career Roadmap Generation

The system generates personalised career development plans using Artificial Intelligence.

Requirements:

- ❖ Accept the user's current skills and target job role.
- ❖ Generate AI-based career roadmaps.
- ❖ Recommend skills, certifications, learning resources, and projects.

- ❖ Display learning milestones and estimated progression.
- ❖ Store generated roadmaps for future access.

4. Mock Interview System

The platform provides AI-based interview practice.

Requirements:

- ❖ Generate interview questions according to the selected job role.
- ❖ Accept user responses.
- ❖ Evaluate responses using Artificial Intelligence.
- ❖ Provide feedback on technical knowledge, communication, and confidence.
- ❖ Store interview history for future review.

5. Job and Learning Recommendation System

The system recommends suitable jobs and learning resources.

Requirements:

- ❖ Recommend jobs based on user profile and skills.
- ❖ Recommend online courses according to identified skill gaps.
- ❖ Allow users to save favourite jobs and courses.
- ❖ Update recommendations based on user activities.

6. AI Chatbot Assistant

The platform includes an AI-powered chatbot for career guidance.

Requirements:

- ❖ Answer career-related questions.
- ❖ Explain resume analysis reports.
- ❖ Suggest learning resources.
- ❖ Guide users through platform features.
- ❖ Maintain conversation context during active sessions.

7. Administration Module

The system provides administrators with tools to manage and monitor the application.

Requirements:

- ❖ Secure administrator login.
- ❖ User account management.

- ❖ Dashboard displaying platform statistics.
- ❖ Notification management.
- ❖ Contact message management.
- ❖ Activity monitoring and reporting.
- ❖ View AI usage statistics.
- ❖ Enable or disable maintenance mode.
- ❖ Manage application cache.

8. Security and Data Management

The system protects user information and application resources.

Requirements:

- ❖ Secure JWT authentication.
- ❖ Role-based authorization.
- ❖ Request validation.
- ❖ API rate limiting.
- ❖ Application logging.
- ❖ Secure storage of user information.
- ❖ Backup and recovery support.

NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the quality attributes and operational characteristics of the proposed system.

1. Performance

The system should provide fast and efficient responses during normal operation.

Requirements:

- ❖ API responses should be completed within acceptable response times.
- ❖ AI-based operations should complete within a reasonable processing time.
- ❖ Pages should load quickly.
- ❖ Support multiple users simultaneously.

2. Security

The system should ensure confidentiality, integrity, and secure access.

Requirements:

- ❖ JWT-based authentication.
- ❖ Password encryption.
- ❖ Request validation.

- ❖ API rate limiting.
- ❖ Secure logging.
- ❖ Protection against unauthorized access.

3. Reliability

The system should continue operating correctly even when unexpected situations occur.

Requirements:

- ❖ Graceful handling of AI service failures.
- ❖ Proper exception handling.
- ❖ Reliable database operations.
- ❖ Stable application performance.

4. Scalability

The application should support future growth.

Requirements:

- ❖ Support increasing numbers of users.
- ❖ Flexible MongoDB database structure.
- ❖ Stateless backend architecture.
- ❖ Easy addition of new modules.

5. Usability

The system should be simple and convenient for users.

Requirements:

- ❖ User-friendly interface.
- ❖ Responsive design for desktop, tablet, and mobile devices.
- ❖ Loading indicators and confirmation messages.
- ❖ Easy navigation.
- ❖ Light and Dark mode support.

TECHNOLOGIES USED

The development of Carvion AI involves modern web technologies, Artificial Intelligence services, databases, and software development tools. The technologies used in the project are described below.

FRONTEND TECHNOLOGIES

Frontend technologies are responsible for developing the user interface and providing an interactive experience for users.

1. React.js (Version 18)

React.js is used to develop the frontend of the application as a Single Page Application (SPA). It provides a component-based architecture that improves code reusability, simplifies user interface development, and enables faster rendering of web pages. React also provides an interactive and responsive user experience.

2. Vite

Vite is used as the frontend build tool and development server. It provides fast project startup, quick module updates during development, and optimized production builds, which improves the overall development process.

3. React Router DOM

React Router DOM is used to implement client-side navigation between different pages of the application without reloading the entire website. It also supports protected routes for authenticated users.

4. Axios

Axios is used to establish communication between the React frontend and the Django backend through REST APIs. It sends HTTP requests and receives responses from the server while supporting JWT authentication.

5. TanStack Query (React Query)

TanStack Query is used for efficient data fetching and server state management. It automatically caches API responses, synchronizes data, and updates the user interface whenever backend data changes.

6. Recharts

Recharts is used to create graphical reports and dashboard visualizations. It

displays analytical information through charts such as bar charts, pie charts, line charts, and area charts.

7. React Icons

React Icons provides lightweight SVG icons that improve the appearance and usability of the application. Icons are used in menus, buttons, dashboards, forms, and navigation components.

8. 1.8 HTML5, CSS3 and JavaScript (Vanilla CSS)

HTML5 provides the structure of web pages, CSS3 is used for designing responsive layouts and animations, while JavaScript adds dynamic functionality to the application. Vanilla CSS is used for custom styling, responsive layouts, dark mode, animations, and overall interface design without relying on external CSS frameworks.

BACKEND TECHNOLOGIES

Backend technologies are responsible for implementing business logic, user authentication, AI integration, and API services.

1. Python (Version 3.11)

Python is the primary programming language used for backend development. It is used to implement application logic, AI integration, document processing, authentication, and API development.

2. Django

Django is the main web framework used to develop the backend application. It provides project organization, authentication support, URL routing, security features, and database management.

3. Django REST Framework (DRF)

Django REST Framework is used to develop RESTful APIs that connect the frontend with the backend. It manages serialization, API requests, permissions, authentication, and response handling.

4. Simple JWT

Simple JWT is used to implement secure user authentication through JSON Web Tokens (JWT). It manages user login sessions, access tokens, refresh tokens, and secure logout operations.

5. MongoEngine ODM

MongoEngine is used as the Object Document Mapper (ODM) between Django and MongoDB. It simplifies the storage and retrieval of user profiles, resumes, AI-generated content, interview data, and career roadmaps.

6. PyMuPDF

PyMuPDF is used to extract text from uploaded PDF resumes before AI analysis.

7. python-docx

python-docx is used to extract text from Microsoft Word (DOCX) resumes.

8. Google Generative AI SDK

The Google Generative AI SDK is used to integrate the Google Gemini 2.5 Flash API with the backend. It enables communication between the application and the AI model for resume analysis, career guidance, interview evaluation, chatbot assistance, and recommendation generation.

9. django-cors-headers

This middleware enables secure communication between the React frontend and Django backend when they are hosted on different domains or ports.

10. Python Logging

Python's built-in logging module is used to record application events, errors, security activities, and system operations into log files for debugging and monitoring.

DATABASE TECHNOLOGIES

1. MongoDB

MongoDB is the primary NoSQL database used to store resumes, user profiles, career roadmaps, interview sessions, job recommendations, learning recommendations, chatbot conversations, and other AI-generated information. Its flexible document structure makes it suitable for storing dynamic data.

2. SQLite

SQLite is used to manage administrator information, authentication data, sessions, and Django system metadata. It provides lightweight and reliable storage for administrative operations.

ARTIFICIAL INTELLIGENCE TECHNOLOGY

1. Google Gemini 2.5 Flash API

Google Gemini 2.5 Flash is the Artificial Intelligence model used throughout the application. It performs resume analysis, ATS score generation, skill gap analysis, career roadmap generation, interview question generation, interview evaluation, chatbot responses, and personalised career recommendations.

DEVELOPMENT TOOLS

1. Visual Studio Code

Used as the primary Integrated Development Environment (IDE) for developing both frontend and backend applications.

2. Git

Git is used for version control to track source code changes and manage project development efficiently.

3. Postman

Postman is used to test REST APIs before integrating them with the frontend application.

4. Npm

npm (Node Package Manager) is used to install, update, and manage frontend libraries and project dependencies.

5. Pip

pip is the package manager used to install and manage Python libraries required for backend development.

6. PowerShell

PowerShell is used during development to execute backend and frontend servers, manage project files, and perform development-related commands on the Windows operating system.

DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves through an information system. It shows the flow of data between external entities, system processes, data stores, and outputs. DFDs help in understanding the overall functioning of the system by clearly representing how information is collected, processed, stored, and transferred within the application.

In the Carvion AI platform, Data Flow Diagrams are used to represent the movement of data between users, administrators, the application modules, the database, and the Google Gemini 2.5 Flash API. They illustrate how user inputs, such as resume files, career preferences, and interview responses, are processed by different modules to generate ATS scores, career roadmaps, job recommendations, learning suggestions, interview feedback, and chatbot responses. The DFDs also show how processed information is stored in the database and retrieved whenever required by the users or administrators.

The following Data Flow Diagrams describe the complete data flow of the Carvion AI system at different levels:

LEVEL 0 – CONTEXT DIAGRAM

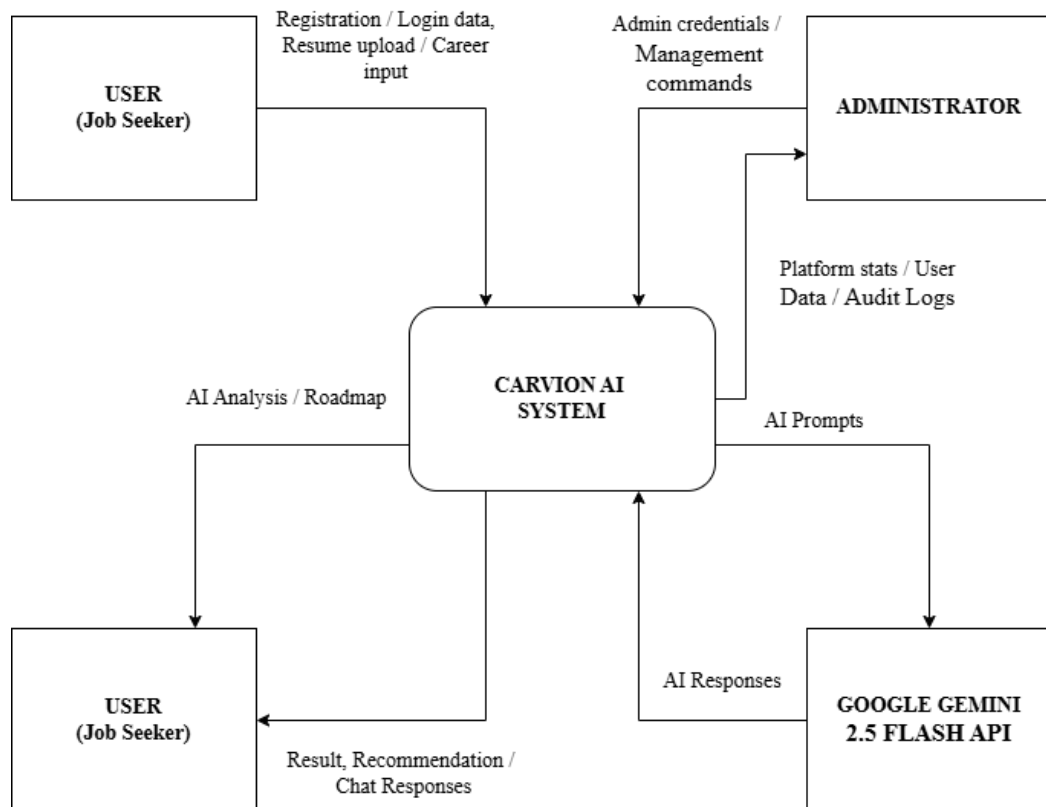


Figure: DFD Level 0 - Context Diagram

The Level 0 Data Flow Diagram (Context Diagram) represents the Carvion AI platform as a single process and illustrates its interaction with external entities. It provides a high-level view of the entire system without showing internal processes or data stores. The purpose of the context diagram is to define the system boundaries and identify the flow of information between the system and its external users or services.

In the Carvion AI platform, the primary external entities include the User, Administrator, and the Google Gemini 2.5 Flash API. Users interact with the system by registering, logging in, uploading resumes, requesting career roadmaps, participating in mock interviews, searching for jobs and learning resources, and communicating with the AI chatbot. Administrators manage user accounts, monitor system activities, generate reports, and maintain the platform through the administration module. The system communicates with the Google Gemini 2.5 Flash API to perform AI-based tasks such as resume analysis, ATS score generation, career roadmap creation, interview evaluation, chatbot assistance, and personalised recommendations.

The Level 0 Context Diagram provides an overall understanding of how data enters and leaves the Carvion AI platform before examining the detailed internal processes in the subsequent DFD levels.

LEVEL 1 – MAIN SYSTEM PROCESSES

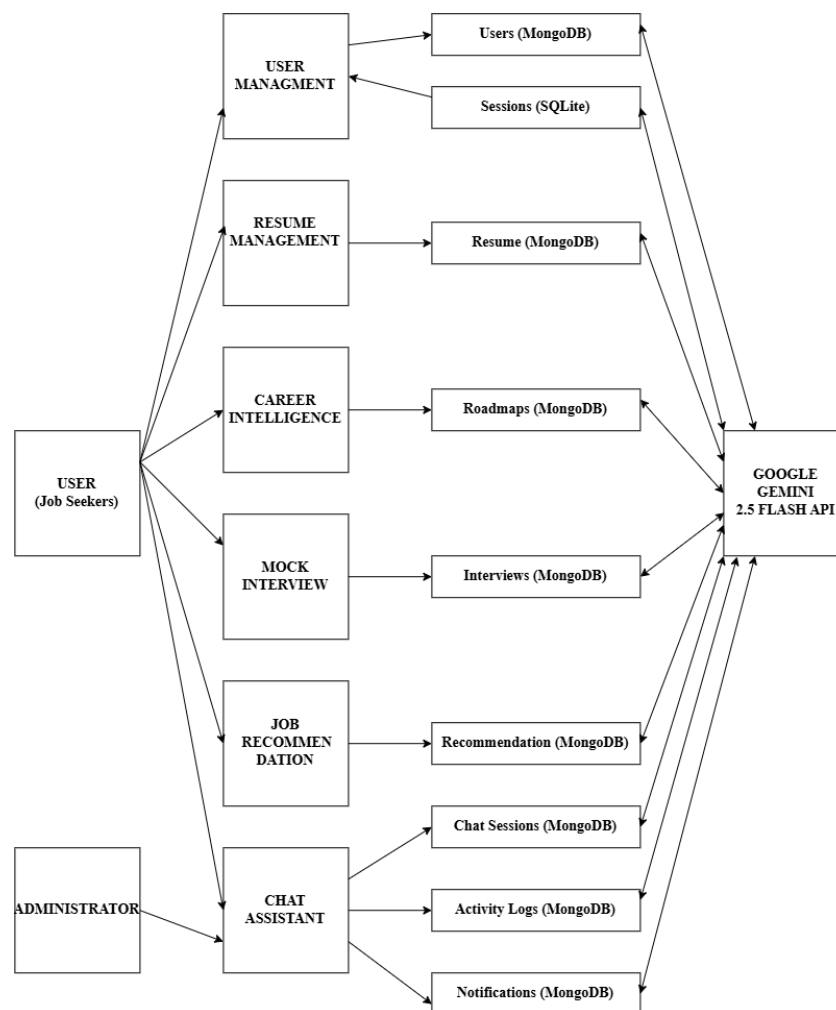


Figure: DFD Level 1 - Main System Processes

The Level 1 Data Flow Diagram (DFD) expands the Level 0 Context Diagram by dividing the Carvion AI platform into its major functional processes. It illustrates how data flows between different system modules, data stores, and external entities. This level provides a detailed view of the core functionalities while showing how information is processed and stored within the system.

The major processes included in the Level 1 DFD are:

1. User Authentication and Account Management

Handles user registration, login, profile management, role-based access control, and secure authentication.

2. Resume Management

Manages resume upload, document parsing, ATS score generation, resume analysis, and storage of resume information.

3. Career Roadmap Generation

Generates personalised career roadmaps based on the user's skills, target job role, and AI analysis.

4. Mock Interview and AI Chatbot

Conducts AI-based mock interviews, evaluates user responses, provides interview feedback, and offers career guidance through the chatbot.

5. Job and Learning Recommendation

Recommends suitable job opportunities and learning resources based on the user's profile, skills, and identified skill gaps.

6. Administration Module

Enables administrators to manage user accounts, monitor platform activities, view reports, manage notifications, and maintain the overall system.

The Level 1 DFD also includes the primary data stores, such as MongoDB, which stores user profiles, resumes, AI-generated roadmaps, interview sessions, recommendations, and chatbot data, and SQLite, which stores administrator information, authentication data, and other system-related records. The diagram also shows the interaction with the Google Gemini 2.5 Flash API, which performs AI-based operations including resume analysis, career roadmap generation, interview evaluation, chatbot responses, and recommendation generation.

The Level 1 Data Flow Diagram provides a clear understanding of how the major functional modules interact with each other and how data flows through the Carvion AI platform to perform its core operations.

LEVEL 2 – PROCESS DECOMPOSITION (RESUME MANAGEMENT)

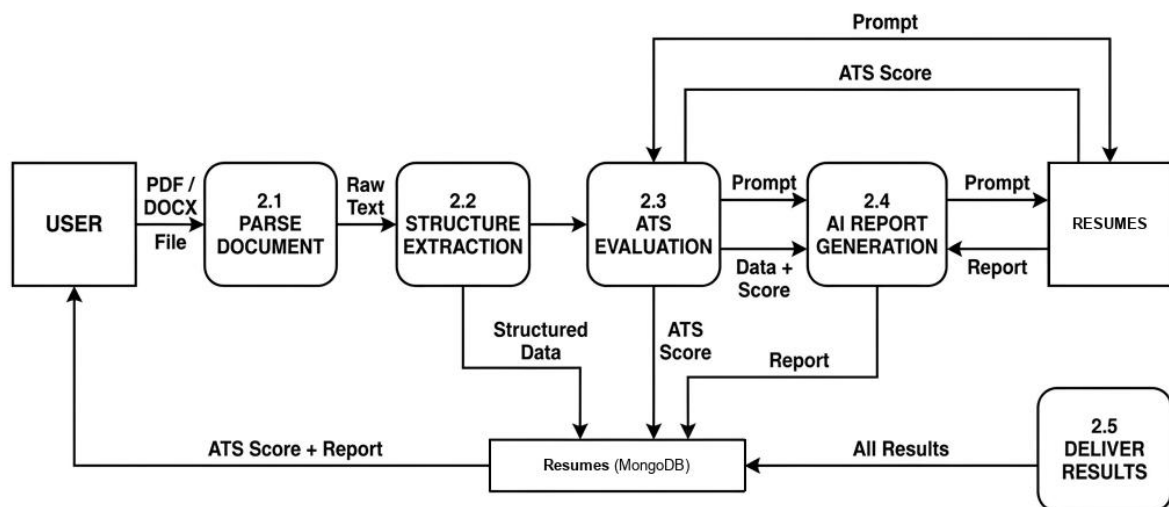


Figure: DFD Level 2 - Resume Management

The Level 2 Data Flow Diagram (DFD) provides a detailed view of the Resume Management process by decomposing Process 2.0 into its internal sub-processes. It illustrates how a resume is uploaded, processed, analysed, stored, and made available to the user. This level helps in understanding the sequence of operations involved in resume management and the flow of data between the user, system modules, database, and the AI service.

The Resume Management process consists of the following sub-processes:

1. Resume Upload

The user uploads a resume in PDF or DOCX format. The system validates the file format and securely stores the uploaded document.

2. Resume Parsing

The uploaded resume is processed using PyMuPDF and python-docx to extract structured information such as personal details, education, work experience, technical skills, projects, and certifications.

3. AI Resume Analysis

The extracted information is sent to the Google Gemini 2.5 Flash API for resume evaluation. The AI analyses the resume, generates an ATS compatibility score, identifies missing skills, and provides recommendations for improvement.

4. Data Storage

The original resume, extracted information, ATS score, and AI-generated analysis are stored in the MongoDB database for future access.

5. Resume Management

The user can view, update, download, or delete previously uploaded resumes and access the corresponding analysis reports whenever required.

The Level 2 DFD clearly illustrates the detailed workflow of the Resume Management module and shows how data moves between the user, processing components, AI service, and database to provide accurate resume analysis and management within the Carvion AI platform.

LEVEL 3 — SUB-PROCESS DETAIL

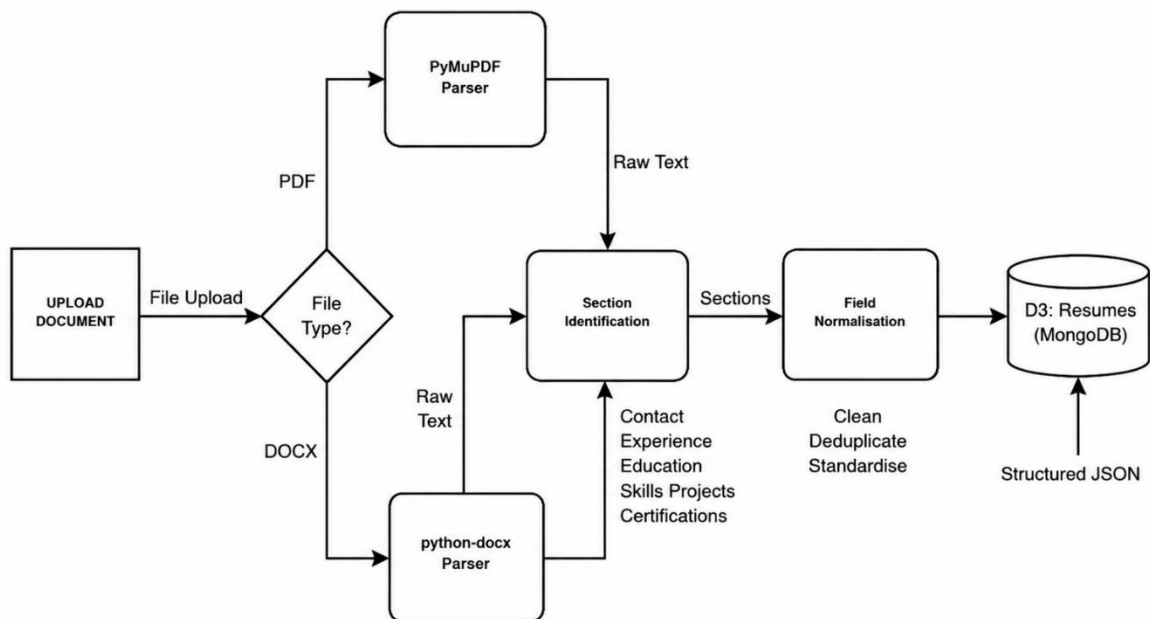


Figure: DFD Level 3 - Document Parsing

The Level 3 Data Flow Diagram (DFD) presents the most detailed view of the Document Parsing and Information Extraction process by further decomposing Processes 2.1 (Resume Upload) and 2.2 (Resume Parsing) into individual processing steps. It illustrates how the uploaded resume is validated, processed, analysed, and converted into structured information before being stored in the database and used for AI-based analysis.

The document parsing and extraction process consists of the following sub-processes:

1. File Upload and Validation

The user uploads a resume in PDF or DOCX format. The system verifies the file format and validates the uploaded document before processing.

2. Text Extraction

The system extracts the textual content from the uploaded resume using PyMuPDF

for PDF files and python-docx for DOCX files.

3. Information Extraction

The extracted text is analysed to identify important information such as personal details, educational qualifications, work experience, technical skills, projects, certifications, and other relevant resume sections.

4. AI Processing

The structured resume information is sent to the Google Gemini 2.5 Flash API for detailed analysis. The AI generates an ATS compatibility score, identifies skill gaps, provides resume improvement suggestions, and prepares additional career-related insights.

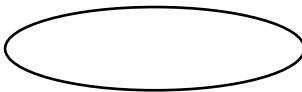
5. Data Storage

The original resume, extracted information, and AI-generated analysis are securely stored in the MongoDB database, allowing users to access their resume history and reports at any time.

6. Result Generation

The processed information, ATS score, extracted skills, and improvement recommendations are returned to the user through the Resume Management module for viewing and further actions.

NOTATIONS USED

NAME	NOTATION	DESCRIPTION
Process		A process transforms incoming dataflow into outgoing dataflow. A process is shown by named circles.
Data flow		Dataflows are pipelines through which packet of information flow. Label the arrows with the name of the data that moves through it.
External entry		External Entities are objects outside the system, with which the system communicates. External Entities are the sources and the destinations of the system's input and output.

ENTITY RELATIONSHIP DIAGRAM

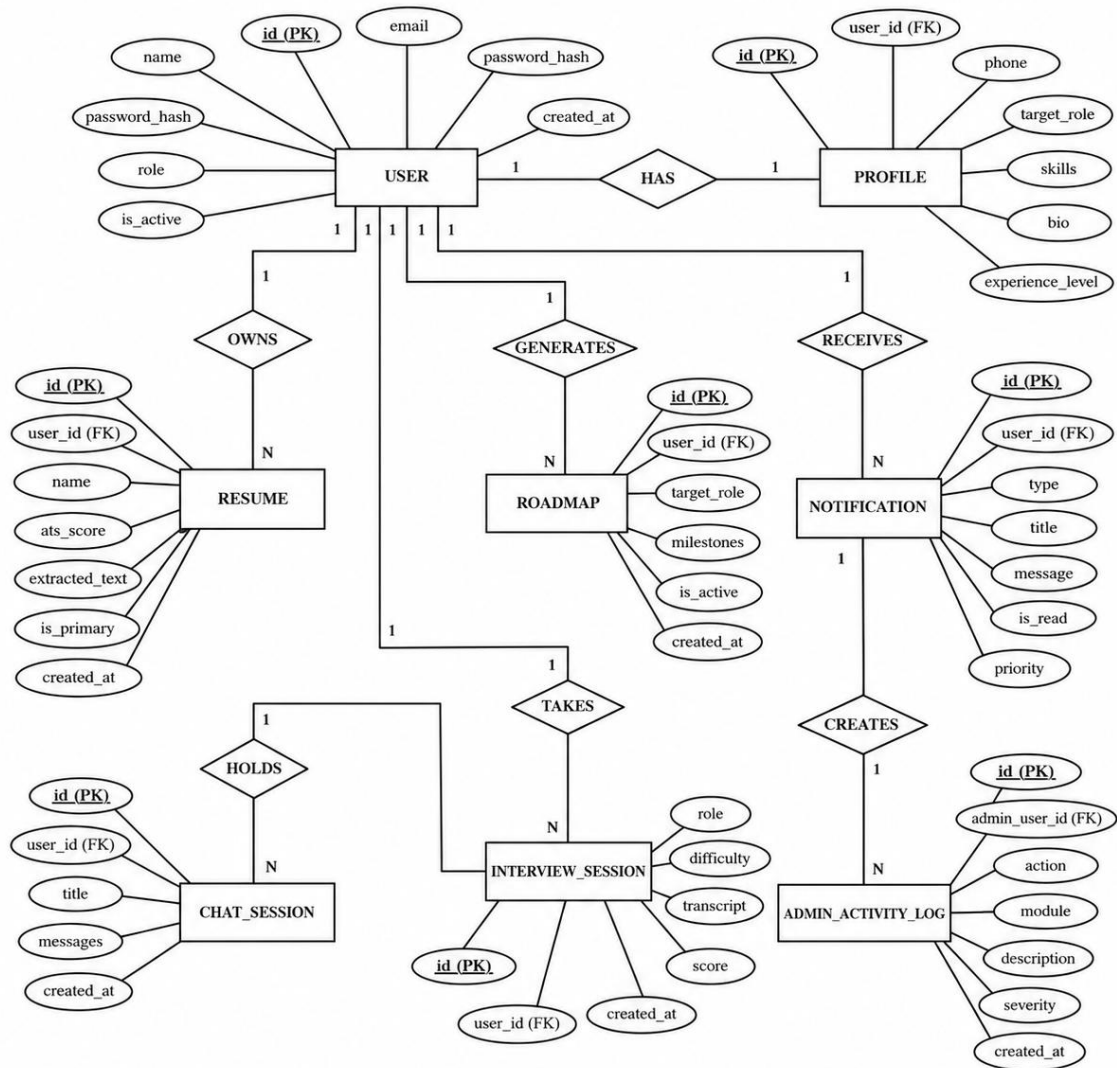


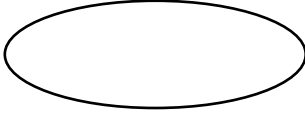
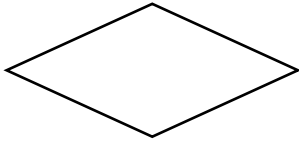
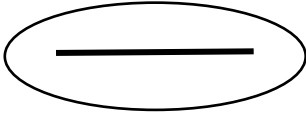
Figure: ER Diagram - Carvion AI

Entity Relationship (ER) Diagram represents the logical database design of the Carvion AI platform. It illustrates the major entities, their attributes, primary keys, foreign keys, and the relationships between them. The ER Diagram provides a clear understanding of how data is organized, stored, and related within the system.

The Carvion AI database consists of several interconnected entities, including User, Administrator, Resume, Career Roadmap, Mock Interview, Job Recommendation, Learning Recommendation, Chatbot Conversation, Notification, Contact Message, and Activity Log. Each entity stores information required for a specific module of the application, while relationships between entities ensure efficient data management and maintain data integrity.

The ER Diagram provides the foundation for database design by defining how entities are connected and how information flows between different modules. It also helps ensure consistency, reduce data redundancy, and maintain the integrity of the database throughout the application.

NOTATIONS USED

NAME	NOTATION	DESCRIPTION
Attribute		The properties of entity can be attribute. It is represented by ellipse.
Relationship		Whenever an attribute of one entity refers to another entity, some relations exists.
Entity		It may be an object with physical existence or conceptual existence.
Link		Lines link attribute to entity sets an entity set relation
Key Attribute		Primary Key: It signifies unique and non null values.

SEQUENCE DIAGRAM

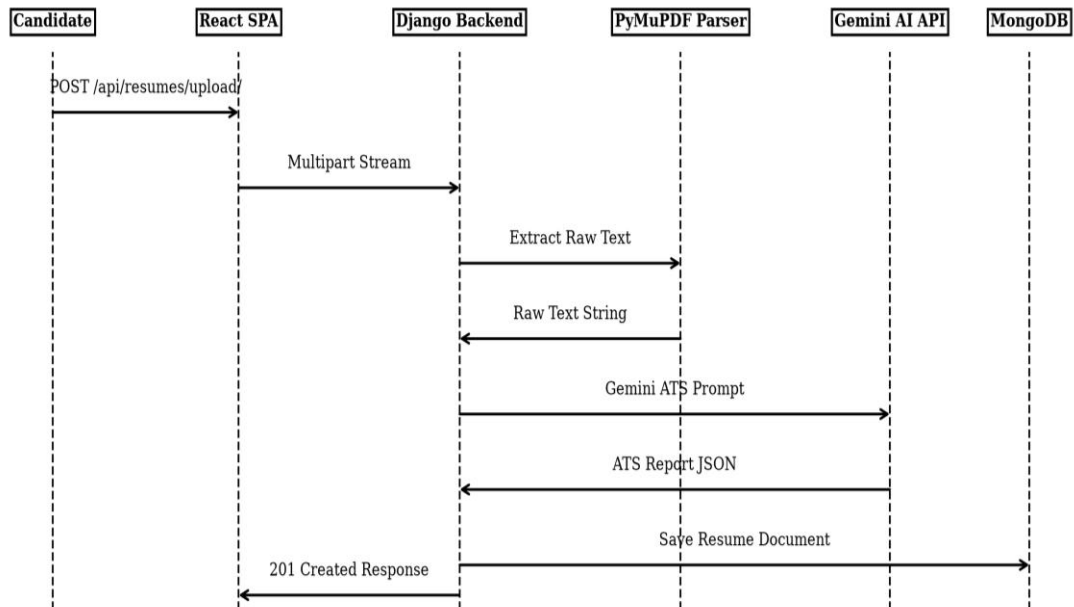


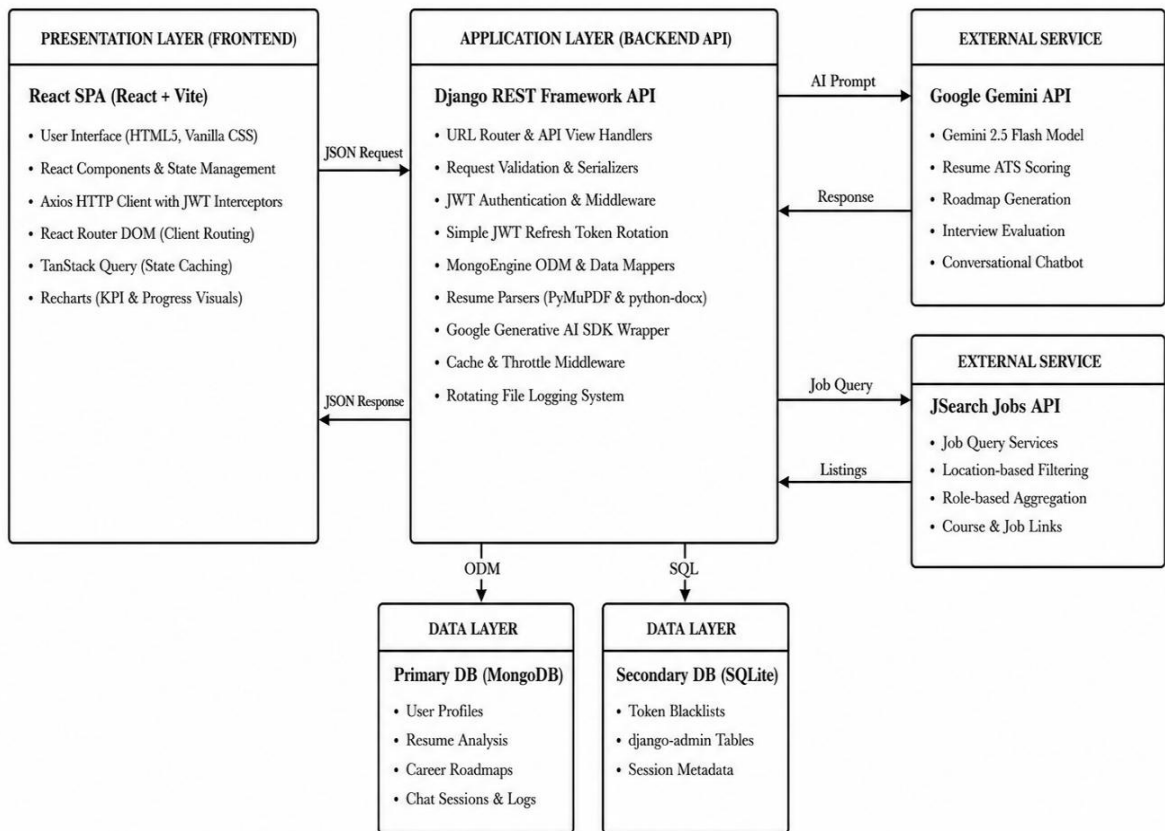
Figure: Sequence Diagram - Carvion AI

The Sequence Diagram illustrates the chronological sequence of interactions between different system components involved in the Resume Upload and Analysis process. It shows how messages are exchanged among the user, frontend application, backend server, document processing module, Artificial Intelligence service, and database to complete the workflow successfully.

In the Carvion AI platform, the sequence begins when the user uploads a resume through the React-based user interface. The request is forwarded to the Django backend, where the uploaded document is processed using PyMuPDF or python-docx to extract the resume content. The extracted information is then sent to the Google Gemini 2.5 Flash API for AI-based analysis. The AI service generates an ATS compatibility score, identifies skill gaps, and provides resume improvement recommendations. Finally, the backend stores the resume and analysis results in the MongoDB database and returns the processed information to the frontend, where the user can view the analysis report.

The Sequence Diagram provides a clear understanding of the order in which system components interact and how requests and responses are exchanged during the Resume Upload and Analysis workflow.

SYSTEM DESIGN



The System Design of the Carvion AI platform describes the overall architecture, major system components, and the interaction between different layers of the application. It provides a structured view of how user requests are processed, how data is managed, and how different technologies work together to deliver the required functionalities. The system follows a three-tier architecture, which separates the presentation layer, application layer, and data layer to improve modularity, maintainability, and scalability.

1. Presentation Layer (Frontend)

The Presentation Layer provides the graphical user interface through which users interact with the system. It is developed using React.js and Vite as a Single Page Application (SPA). This layer manages user authentication, resume upload, dashboard visualization, career roadmap display, mock interviews, job recommendations, chatbot interactions, and administrator operations. Communication with the backend is performed through RESTful APIs using Axios, while Recharts is used to display analytical reports and dashboard charts.

2. Application Layer (Backend)

The Application Layer contains the core business logic of the system and is developed using Python, Django, and Django REST Framework (DRF). It processes user requests, performs authentication and authorization using JSON Web Tokens (JWT), validates input data, manages resume processing, and controls communication with the database and external AI services. The backend also uses PyMuPDF and python-docx to extract

information from uploaded resumes before sending the extracted content for AI-based analysis.

3. Data Layer (Database)

The Data Layer is responsible for storing and managing application data. The system uses a hybrid database approach consisting of MongoDB and SQLite. MongoDB stores user profiles, resumes, career roadmaps, interview sessions, chatbot conversations, job recommendations, learning recommendations, and other AI-generated information. SQLite is used to maintain authentication-related data, administrator information, session details, and Django system metadata required for application management.

4. External Service Integration

The Carvion AI platform integrates external services to provide Artificial Intelligence and job recommendation features. The backend communicates with the Google Gemini 2.5 Flash API to perform resume analysis, ATS score generation, skill gap analysis, career roadmap generation, mock interview evaluation, chatbot assistance, and personalised career guidance. The system also integrates the JSearch API to retrieve job opportunities that match the user's skills, qualifications, and target job role.

DETAILED DESIGN

The Detailed Design phase describes the internal architecture, functional modules, database organisation, and overall workflow of the Carvion AI Career Development Platform. It explains how different components of the system interact to provide AI-powered career development services such as resume analysis, career roadmap generation, mock interviews, job recommendations, and career assistance.

The system is developed using a modular architecture that separates the presentation layer, application layer, database layer, and Artificial Intelligence services. This modular design improves system maintainability, scalability, and simplifies future enhancements.

FRONTEND TECHNOLOGIES

The frontend provides an interactive user interface for accessing all platform features. The following technologies are used:

1. React (Version 18)

Used to develop a component-based Single Page Application (SPA).

2. Vite

Used as the frontend build tool and development server.

3. React Router DOM

Handles client-side routing and protected navigation.

4. TanStack Query (React Query)

Manages server state and efficient data fetching.

5. Recharts

Displays dashboard analytics and graphical reports.

6. Axios

Performs communication between the frontend and backend REST APIs.

7. Vanilla CSS

Used for custom styling, responsive layouts, animations, and user interface design.

BACKEND TECHNOLOGIES

The backend manages business logic, user authentication, resume processing, and communication with AI services.

The backend is developed using:

1. Python (Version 3.11)

Primary programming language.

2. Django (Version 4.x)

Web application framework.

3. Django REST Framework

Provides RESTful API services.

4. Simple JWT

Implements secure JSON Web Token authentication.

5. MongoEngine

Object Document Mapper (ODM) for MongoDB.

6. PyMuPDF (fitz)

Extracts text from PDF resumes.

7. python-docx

Reads and processes Microsoft Word resumes.

8. Google Generative AI SDK

Integrates the Google Gemini 2.5 Flash API for AI-powered career analysis.

DATABASE TECHNOLOGIES

The platform uses two databases for efficient data management.

1. MongoDB

serves as the primary database for storing user profiles, resumes, career roadmaps, interview sessions, chatbot conversations, recommendations, and AI-generated results.

2. SQLite

is used as the secondary database for Django administrative data, authentication-

related information, and session management.

MAJOR MODULES OF THE SYSTEM

1. Authentication Module

The Authentication Module manages user registration, login, logout, and secure session management. During registration, user information is validated, passwords are securely encrypted, and user accounts are created. During login, the system verifies user credentials and generates JSON Web Tokens (JWT) for secure authentication.

The module also implements Role-Based Access Control (RBAC), allowing administrators and normal users to access only their respective functionalities. Secure authentication ensures that only authorised users can access protected resources within the platform.

2. Dashboard Module

The Dashboard Module provides users with a personalised overview of their activities within the platform. It displays resume statistics, ATS scores, career roadmap progress, interview performance, notifications, and profile completion status.

The dashboard retrieves data from multiple backend services and presents the information using interactive charts and progress indicators, enabling users to monitor their career development effectively.

3. Resume Management Module

The Resume Management Module manages the complete resume lifecycle, including uploading, parsing, AI-based analysis, storage, and retrieval.

Users can upload resumes in PDF or DOCX format. The system extracts textual information using PyMuPDF or python-docx, identifies important resume sections, and sends the extracted information to the Google Gemini 2.5 Flash API for ATS evaluation.

The AI generates an ATS score, identifies missing skills, and provides recommendations to improve resume quality. All resume information and analysis reports are securely stored in MongoDB for future reference.

4. Career Roadmap Module

The Career Roadmap Module generates personalised learning paths according to the user's current skills and target career.

Based on the user's profile information, the system communicates with the Google Gemini 2.5 Flash API to generate milestone-based career roadmaps. Each roadmap includes learning objectives, required skills, recommended learning resources, and estimated completion timelines.

Users can monitor their progress by marking milestones as completed.

5. Mock Interview Module

The Mock Interview Module provides an AI-assisted interview preparation environment.

Users select a target job role and interview category. The Google Gemini 2.5 Flash API generates interview questions according to the selected role. After users submit their responses, the AI evaluates each answer and provides detailed feedback including strengths, weaknesses, communication quality, and improvement suggestions.

All interview sessions are stored in MongoDB for future review.

6. Recommendations Module

The Recommendations Module provides personalised job opportunities and learning resources.

Job recommendations are retrieved using the JSearch API based on the user's skills, preferred location, and career goals. Learning recommendations are generated according to the skill gaps identified during resume analysis.

Users can save jobs and courses for future reference.

7. AI Chatbot Module

The AI Chatbot Module provides conversational career guidance throughout the platform.

Users can ask questions related to resumes, interview preparation, career planning, roadmap milestones, and job searching. The chatbot is powered by the Google Gemini 2.5 Flash API and maintains conversation history to provide context-aware responses.

8. Admin Console Module

The Admin Console Module enables administrators to monitor and manage the entire platform.

Administrators can manage users, review contact messages, send notifications, monitor system statistics, and perform administrative operations through a secure dashboard.

The module also maintains activity logs for auditing administrative actions.

9. Database Architecture

The Carvion AI platform adopts a hybrid database architecture consisting of MongoDB and SQLite.

MongoDB stores all major application data including user profiles, resumes, career roadmaps, interview sessions, chatbot conversations, job recommendations, learning recommendations, and AI-generated reports. Its document-oriented structure provides flexibility for handling varying data formats. SQLite is used to manage Django system

data, authentication metadata, session information, and administrative records. This separation improves system organisation and enhances overall performance.

10. Security Architecture

The platform incorporates multiple security mechanisms to protect user information and ensure secure system operation.

The implemented security features include:

- ❖ JWT Authentication for secure API access.
- ❖ Role-Based Access Control (RBAC) to restrict administrative functions.
- ❖ Password Encryption to securely store user credentials.
- ❖ Input Validation to prevent invalid or malicious data.
- ❖ API Rate Limiting to reduce misuse of backend services.
- ❖ Application Logging for monitoring security and system events.
- ❖ Cross-Origin Resource Sharing (CORS) configuration to allow requests only from authorised frontend applications.

These security mechanisms improve data protection, maintain system integrity, and provide secure communication between users and the Carvion AI platform.

DATABASE DESIGN

Database design is an important component of the Carvion AI Career Development Platform. It is responsible for storing, managing, and retrieving all information related to users, resumes, career roadmaps, interview sessions, job recommendations, chatbot conversations, and administrative records.

The platform primarily uses MongoDB, a document-oriented NoSQL database, with MongoEngine as the Object Document Mapper (ODM) for communication between the Django backend and the database. SQLite is also used to store Django administrative data, authentication information, and session metadata.

The database is designed to ensure efficient data storage, consistency, security, and reliable retrieval of information. Multiple interconnected collections are used, where each collection represents a specific module of the system. This structured database design supports the various functionalities of the platform while allowing future enhancements with minimal modifications.

MAJOR DATABASE TABLES

Users collection

Field Name	Data Type	Key	Description
_id	ObjectId	PK	Unique auto-generated document identifier
email	String(255)	Unique	User email address used for authentication
username	String(150)	Unique	Optional display username
password_hash	String(255)	—	Bcrypt-hashed password. Null for OAuth users.
name	String(255)	—	Full display name of the user
role	String	—	Account role: 'standard' or 'admin'
is_active	Boolean	—	Whether the account is active and can log in
is_pending_deletion	Boolean	—	Flags account scheduled for deletion
scheduled_deletion_date	DateTime	—	Date the account will be permanently deleted
created_at	DateTime	—	Account creation timestamp (UTC)
updated_at	DateTime	—	Last account update timestamp (UTC)

Resumes collection

Field Name	Data Type	Key	Description
id	ObjectId	PK	Unique document identifier
user	ObjectId	FK → users	Reference to the owning user document
name	String(255)	—	Friendly identifier for the resume
file_name	String(255)	—	Original uploaded file name
file_path	String	—	Server-side path to the stored file
extracted_text	String	—	Raw text extracted by PyMuPDF or python-docx
structured_data	Object	—	Parsed resume fields: experience, education, skills, projects, certifications
ats_score	Integer	—	ATS compatibility score (0–100) generated by Gemini
analysis_report	Object	—	Gemini improvement report as a structured JSON object
downloads_count	Integer	—	Number of times the resume has been downloaded
is_primary	Boolean	—	Whether this is the user's primary resume
created_at	DateTime	—	Upload/creation timestamp (UTC)
updated_at	DateTime	—	Last update timestamp (UTC)

Roadmaps collection

Field Name	Data Type	Key	Description
id	ObjectId	PK	Unique document identifier
user	ObjectId	FK → users	Reference to the owning user document
target_role	String(255)	—	The career role this roadmap is built for
milestones	List[Object]	—	Array of milestone nodes (title, description, timeframe, skills, references, is_completed)
is_active	Boolean	—	Whether this is the user's currently active roadmap
is_system_generated	Boolean	—	True if auto-generated by the system on profile update
profile_state_hash	String	—	Hash of the profile at the time of generation, used to detect stale roadmaps
schema_version	Integer	—	Version number for milestone schema migrations
created_at	DateTime	—	Roadmap creation timestamp (UTC)
updated_at	DateTime	—	Last update timestamp (UTC)

Admin_activity_logs collection

Field Name	Data Type	Key	Description
id	ObjectId	PK	Unique document identifier
admin_user	ObjectId	FK → users	Reference to the administrator who performed the action
action	String(255)	—	Short label describing the action performed
module	String(100)	—	Admin Console module where the action occurred
target_record	String(255)	—	Identifier of the record affected by the action
description	String(1000)	—	Human-readable description of the action
status	String	—	Outcome: success or failed
severity	String	—	Severity level: INFO, WARNING, or CRITICAL
ip_address	String	—	IP address of the administrator at the time of action
user_agent	String	—	Browser or client user agent string
request_id	String	—	Unique identifier for the HTTP request
created_at	DateTime	—	Log entry creation timestamp (UTC)

Mock_tests collection

Field Name	Data Type	Key	Description
id	ObjectId	PK	Unique document identifier
user	ObjectId	FK → users	Reference to the owning user document
domain	String(100)	—	Subject domain of the test (e.g., React, Python)
difficulty	String	—	Difficulty level: Easy, Medium, or Hard
category	String	—	Test type: MCQ, Technical, Coding, Aptitude, HR
questions	List[Object]	—	Array of question objects with options and correct answers
created_at	DateTime	—	Test generation timestamp (UTC)

TESTING

Testing is an essential phase of software development that ensures the application functions correctly, securely, and efficiently. It helps identify software defects, verify system functionality, improve performance, and ensure that the developed application satisfies both functional and non-functional requirements.

In the Carvion AI Career Development Platform, various testing techniques were performed to evaluate the functionality of individual modules as well as the overall system. These testing activities helped ensure the reliability, security, and usability of the platform.

UNIT TESTING

Unit Testing is performed to verify the functionality of individual modules independently. Each module is tested separately to ensure that it performs its intended operation correctly before integration with other modules.

Sample Test Cases

1. Verified that duplicate email addresses are not allowed during user registration.
2. Tested successful login and JWT token generation.
3. Verified text extraction from PDF and DOCX resume files.
4. Confirmed that ATS scores are generated successfully for uploaded resumes.
5. Tested JWT validation for protected API requests.
6. Verified CRUD operations on MongoDB collections.

INTEGRATION TESTING

Integration Testing verifies that different modules communicate correctly after integration.

In Carvion AI, the frontend, backend, database, and AI services were integrated and tested together to ensure smooth data exchange and workflow execution.

Integration Test Cases

1. Verified complete resume upload and analysis workflow.
2. Tested communication between React frontend and Django REST API.
3. Confirmed successful interaction with the Google Gemini 2.5 Flash API.
4. Verified storage and retrieval of data from MongoDB.
5. Tested notification delivery from the Admin Console to users.
6. Verified dashboard updates after backend data modifications.

SYSTEM TESTING

System Testing evaluates the complete Carvion AI platform to ensure that all modules work together according to the project requirements.

The entire application was tested under normal operating conditions to verify overall functionality.

System Test Cases

1. User Registration and Login
2. Profile Management
3. Resume Upload and ATS Analysis
4. Career Roadmap Generation
5. Mock Interview Process
6. Job and Course Recommendations
7. AI Chatbot Functionality
8. Admin Console Operations
9. Notification Management
10. System Configuration

SECURITY TESTING

Security Testing ensures that the application protects user information and prevents unauthorised access.

The Carvion AI platform implements several security mechanisms that were tested during development.

Security Test Cases

1. JWT authentication validation.
2. Role-Based Access Control (RBAC).
3. Password encryption.
4. Access token expiry and refresh mechanism.
5. Input validation and sanitisation.
6. API rate limiting.
7. Session management after logout.
8. Protection of administrator resources.
9. The testing also verified that unauthorised users cannot access protected APIs and that sensitive information is never exposed through API responses.

PERFORMANCE TESTING

Performance Testing evaluates the responsiveness and efficiency of the application under different operating conditions.

The performance of the Carvion AI platform was tested for response time, database operations, AI processing, and concurrent request handling.

Performance Test Cases

1. API response time.
2. Resume analysis execution time.
3. Career roadmap generation time.
4. AI interview evaluation response time.
5. Frontend page loading speed.
6. MongoDB query performance.
7. Multiple user request handling.

USER ACCEPTANCE TESTING (UAT)

User Acceptance Testing (UAT) is the final phase of testing, where end users evaluate the application to determine whether it satisfies their requirements and expectations.

Users interacted with the platform by performing various career development activities and provided feedback regarding system usability, AI-generated results, and overall user experience.

User Acceptance Test Cases

1. Registration and Login
2. Resume Upload and Analysis
3. Career Roadmap Generation
4. Mock Interview
5. Job Recommendations
6. Course Recommendations
7. AI Chatbot
8. Dashboard Navigation
9. Notification System
10. User feedback was used to identify usability improvements and enhance the overall functionality of the platform before final deployment.

TEST CASES

The following test cases cover all major modules of the Carvion AI platform. Each test case specifies the test ID, module, description, input data, expected result, actual result, and pass/fail status.

Module 1 — User Authentication

TC-AUTH-01	Register with valid credentials	Valid name, email, strong password	Account created, JWT token issued, 201 response	Account created successfully	Pass
TC-AUTH-02	Register with duplicate email	Email already in database	400 error: "Email already exists"	Duplicate rejected with error message	Pass
TC-AUTH-03	Register with missing required fields	Empty name or email	400 validation error returned	Validation error returned	Pass
TC-AUTH-04	Login with valid credentials	Correct email and password	Access token and refresh token issued, 200 response	Tokens issued and user logged in	Pass
TC-AUTH-05	Login with incorrect password	Valid email, wrong password	401 Unauthorised: "Invalid credentials"	Login rejected with 401 error	Pass

Module 2 — User Profile

TC-PROF-01	Create profile for new user	Valid phone, target_role, skills, bio	Profile document created in MongoDB, 201 response	Profile created successfully	Pass
TC-PROF-02	Retrieve profile for authenticated user	Valid JWT token	Profile data returned, 200 response	Profile data returned correctly	Pass
TC-PROF-03	Update profile with new target role	Updated target_role and skills	Profile updated in MongoDB, 200 response	Profile updated successfully	Pass
TC-PROF-04	Access another user's profile	Different user's ID in request	403 Forbidden — access denied	Access blocked	Pass

Module 3 — Resume Management

TC-RES-01	Upload valid PDF resume	PDF file within size limit	Document parsed, text extracted, ATS score generated, resume saved to MongoDB	Resume processed and saved	Pass
TC-RES-02	Upload valid DOCX resume	DOCX file within size limit	Document parsed, text extracted, ATS score generated, resume saved to MongoDB	Resume processed and saved	Pass
TC-RES-03	Upload unsupported file format	.txt or .jpg file	400 error: unsupported file type	Upload rejected with error	Pass
TC-RES-04	Upload empty or corrupted file	Zero-byte or corrupted PDF	400 error: file could not be parsed	Upload rejected with error	Pass
TC-RES-05	Retrieve resume list for user	Valid JWT, authenticated user	List of all user's resumes returned with names and ATS scores	Resume list returned correctly	Pass

Module 4 — Career Roadmap

TC-ROAD-01	Generate roadmap for a target role	User profile with target_role set	Milestone-based roadmap created and saved to MongoDB	Roadmap generated and saved	Pass
TC-ROAD-02	Retrieve active roadmap	Valid JWT, user has active roadmap	Active roadmap with milestones returned	Roadmap returned correctly	Pass
TC-ROAD-03	Mark a milestone as completed	Valid milestone ID	Milestone is_completed set to true, progress updated	Milestone marked complete	Pass

TC-ROAD-04	Generate roadmap with no profile data	User with empty profile	400 error or AI generates generic roadmap for given role	Handled gracefully	Pass
TC-ROAD-05	Request roadmap for new role	Different target_role input	New roadmap generated for the new role	New roadmap created	Pass

Module 5 — Mock Interview

TC-INT-01	Start interview session with valid role	Target role, difficulty, category	Interview session created, questions generated by Gemini, 201 response	Session started with questions	Pass
TC-INT-02	Submit answer to interview question	Question ID, answer text	Answer evaluated by Gemini, structured feedback returned with score	Feedback returned with scores	Pass
TC-INT-03	Submit empty answer	Question ID, empty string	400 error: answer cannot be empty	Empty answer rejected	Pass
TC-INT-04	Retrieve interview session history	Valid JWT	List of past sessions with scores and dates returned	History returned correctly	Pass
TC-INT-05	Retrieve session with invalid ID	Non-existent session_id	404 Not Found response	404 returned	Pass

Module 6 — AI Chatbot

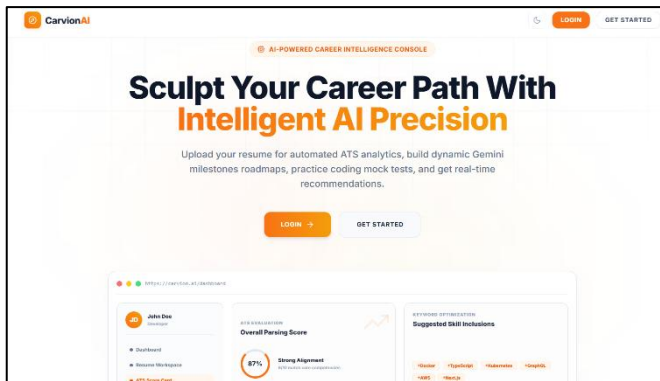
TC-CHAT-01	Send a career-related message	"How do I become a data scientist?"	AI chatbot returns relevant career guidance response	Relevant response returned	Pass
TC-CHAT-02	Send empty message	Empty string in message field	400 error: message cannot be empty	Empty message rejected	Pass

TC-CHAT-03	Retrieve all chat sessions for user	Valid JWT	List of all sessions with titles and timestamps returned	Sessions listed correctly	Pass
TC-CHAT-04	Retrieve specific session messages	Valid session_id	Full message history of session returned	Messages returned correctly	Pass
TC-CHAT-05	Delete a chat session	Valid session_id	Session soft-deleted from MongoDB, 204 response	Session deleted	Pass

Module 7 — Admin Console

TC-ADMIN-01	Login as administrator	Admin email and password	Admin JWT issued with role = admin	Admin session started	Pass
TC-ADMIN-02	Access admin dashboard as admin	Admin JWT	Real-time KPI data returned — user count, resume count, AI usage	Dashboard data returned	Pass
TC-ADMIN-03	Access admin dashboard as standard user	Standard user JWT	403 Forbidden	Access blocked	Pass
TC-ADMIN-04	List all platform users	Admin JWT, search and filter params	Paginated list of all user accounts returned	Users listed with filters applied	Pass
TC-ADMIN-05	Deactivate a user account	Valid user_id, admin JWT	User is_active set to false, user cannot log in	Account deactivated	Pass

SCREENSHOTS

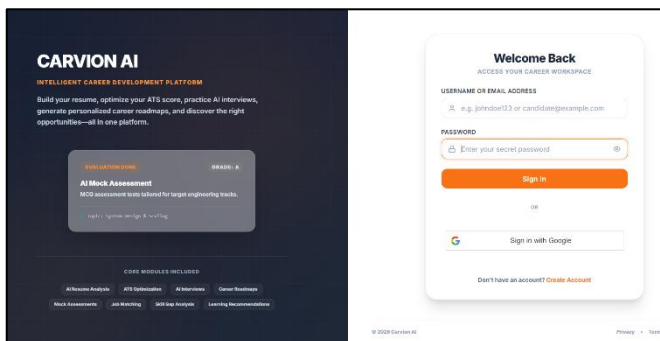
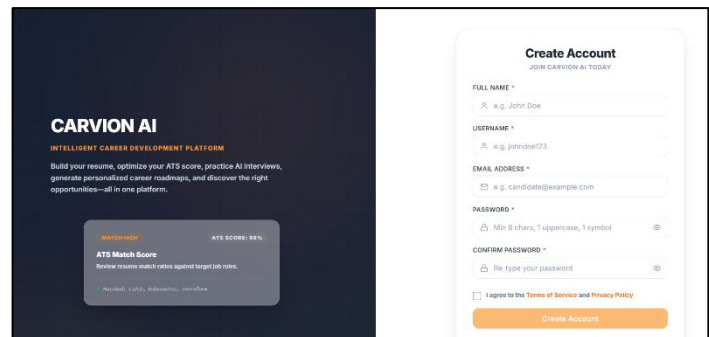


LANDING PAGE

The landing page introduces the Carvion AI platform and highlights its AI-powered career development features. It provides users with options to register or log in and access personalized career guidance services.

REGISTRATION PAGE

The registration page enables new users to create an account by providing the required personal and authentication details. It securely stores user information and grants access to the Carvion AI platform after successful registration.

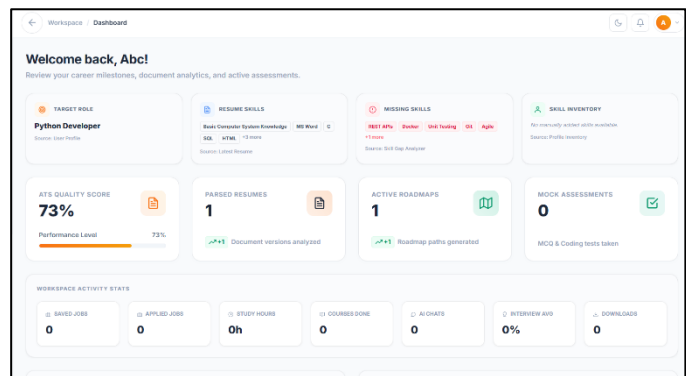


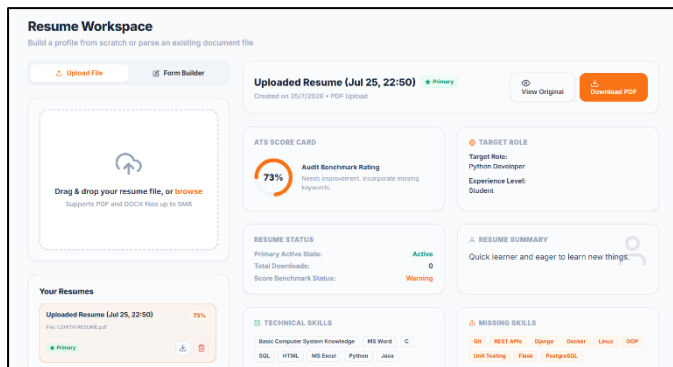
LOGIN PAGE

The login page allows registered users to securely access the Carvion AI platform using their authentication credentials. It ensures authorized access to personalized career analysis, recommendations, and user-specific features.

DASHBOARD

The dashboard provides users with a centralized view of their career progress, resume insights, and personalized recommendations. It enables quick access to all major modules and displays important analytics for effective career planning.



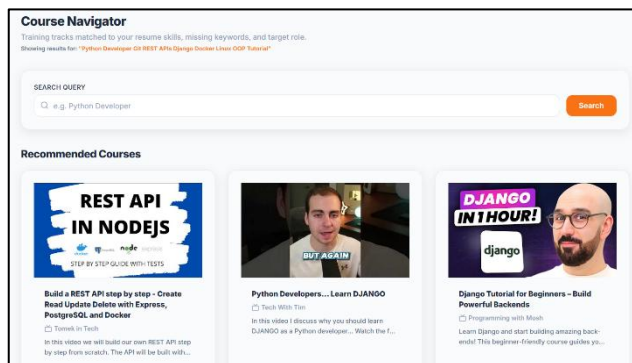
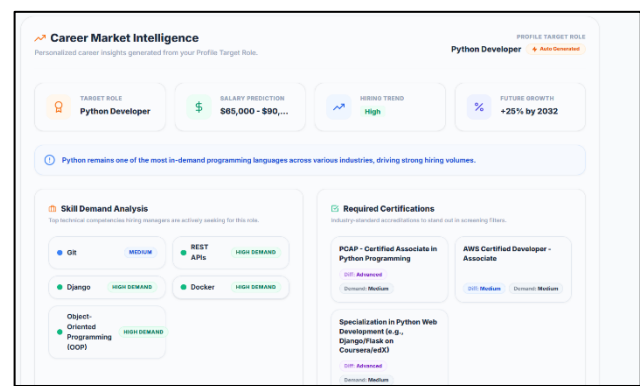


RESUME WORKSPACE

The Resume Workspace enables users to upload, manage, and analyze resumes for AI-based skill extraction and ATS evaluation. It provides a centralized interface for resume optimization and career profile management.

CAREER INSIGHTS

The Career Insights page provides AI-generated analysis of the user's skills, career readiness, and improvement areas. It helps users make informed career decisions through personalized recommendations and detailed performance insights.

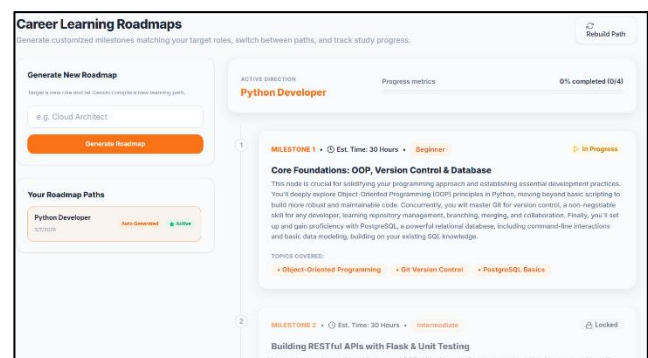


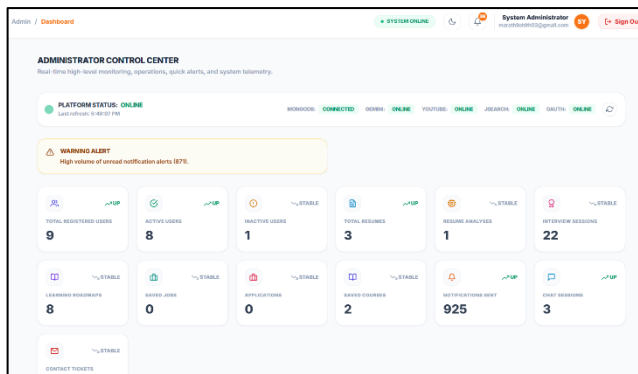
COURSE NAVIGATOR

The Course Navigator recommends personalized learning resources and certification courses based on the user's skill gaps and career goals. It helps users follow a structured learning path to improve their technical and professional competencies.

LEARNING ROADMAPS

The Career Roadmaps page generates a personalized step-by-step learning and career progression plan based on the user's skills and target role. It guides users toward achieving their career objectives through structured milestones and recommendations.



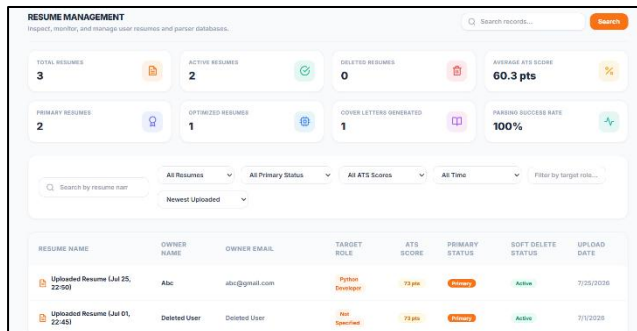
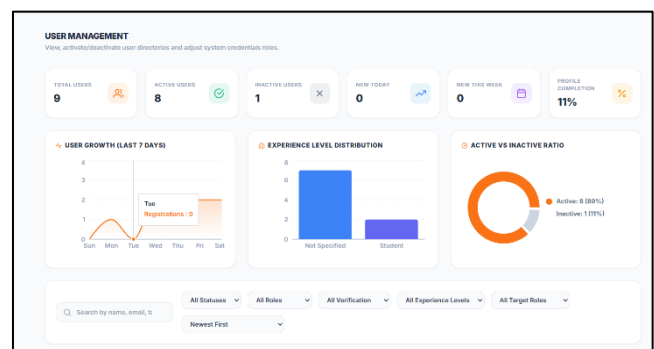


ADMIN DASHBOARD

The Admin Dashboard provides administrators with a centralized interface to monitor users, manage platform activities, and oversee system performance. It enables efficient administration through analytics, user management, and platform monitoring features.

USER MANAGEMENT

The User Management page enables administrators to view, manage, and monitor registered user accounts within the Carvion AI platform. It provides controls for maintaining user information and ensuring efficient platform administration.

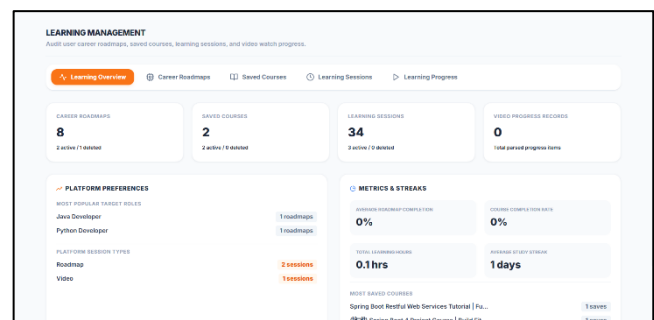


RESUME MANAGEMENT

The Resume Management page enables administrators to view, manage, and monitor all uploaded resumes within the platform. It supports efficient resume administration and ensures proper maintenance of user resume records.

LEARNING MANAGEMENT

The Admin Dashboard provides administrators with a centralized interface to monitor users, manage platform activities, and oversee system performance. It enables efficient administration through analytics, user management, and platform monitoring features.



RESULT AND DISCUSSION

The Carvion AI Career Development Platform was successfully designed, developed, and tested according to the project objectives. The system demonstrates the effective integration of Artificial Intelligence with modern web technologies to provide resume analysis, career guidance, interview preparation, job recommendations, and personalised learning support through a single platform.

The application integrates React, Django REST Framework, MongoDB, SQLite, Google Gemini 2.5 Flash API, and the JSearch API. All major modules, including user authentication, resume management, career roadmap generation, mock interviews, job recommendations, AI chatbot, and administrator functions, operated successfully during testing.

RESULT

1. System Performance

The performance of the platform was evaluated based on response time and overall system efficiency.

- ❖ Resume parsing and AI-based analysis were completed within 8–12 seconds.
- ❖ Frontend page loading time remained below 2.5 seconds under normal network conditions.
- ❖ Job recommendations using the JSearch API were delivered in approximately 1.5 seconds through caching.
- ❖ MongoDB provided efficient storage and retrieval of application data.

The evaluation indicates that the platform delivers satisfactory performance for normal user operations.

2. AI-Based Functional Evaluation

The AI features were evaluated based on the quality and consistency of their outputs.

- ❖ The Google Gemini 2.5 Flash API successfully generated personalised career roadmaps.
- ❖ The resume analysis module produced ATS scores, identified skill gaps, and suggested improvements.
- ❖ The mock interview module evaluated responses and generated structured feedback.
- ❖ The AI chatbot provided relevant career guidance and platform assistance.

The results confirm that the AI components effectively support the core functionalities of the system.

3. Administrative Evaluation

The Administrator Module functioned successfully during testing. It enabled efficient management of users, notifications, contact messages, and system activities. The

administrative dashboard displayed important platform information and supported effective monitoring through a centralized interface.

DISCUSSION

The implementation and testing results show that the Carvion AI Career Development Platform successfully achieved the objectives of the project. The integrated modules work together to provide resume analysis, career roadmap generation, interview preparation, job recommendations, and AI-assisted career guidance within a single application.

The modular architecture, secure authentication mechanism, and database design contribute to the reliability, maintainability, and efficiency of the platform. Overall, the testing results demonstrate that the system satisfies the functional and non-functional requirements specified for the project.

LIMITATIONS OF THE PROPOSED SYSTEM

Every software system has certain limitations due to technology constraints, development time, resource availability, and external service dependencies. The following limitations have been identified in the current version of the Carvion AI Career Development Platform.

1. Dependency on External AI API

The platform depends on the Google Gemini 2.5 Flash API for AI-powered functionalities such as resume analysis, ATS score generation, career roadmap creation, mock interview evaluation, and chatbot responses. If the API is unavailable, exceeds usage limits, or experiences service interruptions, these AI features cannot function until the service becomes available again.

2. AI Response Quality and Consistency

The quality of AI-generated outputs depends on the capabilities of the Google Gemini model. In some situations, especially for specialised technical domains or uncommon job roles, the generated responses may be generic or less accurate. The platform currently relies on the AI-generated output without automatic quality verification.

3. Resume File Format Restrictions

The Resume Management Module currently supports only PDF and DOCX file formats. Other document formats such as DOC, RTF, and ODT are not supported. In addition, scanned image-based PDF resumes cannot be processed accurately because the system extracts only machine-readable text.

4. Absence of Real-Time Communication

The platform does not implement real-time communication technologies such as WebSockets or Server-Sent Events. Notifications, dashboard information, and other dynamic content are updated only when users refresh the page or perform a new request.

5. Single Language Support

The current version of the platform supports only the English language. AI-generated resume analysis, interview questions, career roadmaps, and chatbot responses are produced only in English, limiting accessibility for users who prefer other languages.

6. No Native Mobile Application

Carvion AI is developed as a responsive web application. Although it can be accessed using mobile web browsers, there is currently no dedicated Android or iOS mobile application. Some features may provide a better user experience on desktop or laptop devices.

7. Limited Recommendation Personalisation

The recommendation system mainly considers the user's profile, selected skills, and

target job role. It does not currently use advanced machine learning techniques such as collaborative filtering, behavioural analysis, or recommendation models that continuously learn from user activities.

8. No Email Notification Service

Notifications are delivered only within the application. Users do not receive email alerts for important events such as completed resume analysis, administrator notifications, or account-related activities. Users must log into the platform to view notifications.

9. No Offline Support

The platform requires an active internet connection to access all services. Since AI processing, authentication, database operations, and external APIs require online connectivity, the system cannot be used in offline mode.

10. Limited Job Data Sources

The job recommendation module currently depends on the JSearch API to retrieve job listings. During the development of the project, attempts were made to integrate job data from major job portals such as LinkedIn, Naukri, and Indeed. However, due to API access restrictions, authentication requirements, usage limitations, and the absence of freely available public APIs, direct integration with these platforms could not be implemented. As a result, the platform cannot aggregate job opportunities from multiple job portals, which limits the variety and coverage of recommended job listings.

11. Limited Learning Resource Sources

The learning recommendation module currently provides learning resources primarily through YouTube. Integration with popular online learning platforms such as Coursera, Udemy, edX, and LinkedIn Learning could not be implemented because these platforms do not provide freely accessible public APIs or require commercial API subscriptions. Consequently, the platform is unable to recommend courses from multiple learning providers, resulting in a more limited selection of learning resources.

12. Limited Support for Non-IT Career Domains

The current version of the Carvion AI Career Development Platform is primarily designed for Software and Information Technology (IT) careers. The resume analysis, career roadmaps, interview preparation, job recommendations, and learning resources are mainly focused on IT-related roles and technical skills. As a result, the platform provides limited support for users from non-IT domains such as Mechanical, Civil, Electrical, Electronics, Healthcare, Finance, Marketing, Law, Commerce, and other business sectors. Future versions can expand the knowledge base, AI prompts, job sources, and learning recommendations to support a wider range of academic disciplines and professional careers.

FUTURE SCOPE AND ENHANCEMENTS

The Carvion AI Career Development Platform provides a strong foundation for AI-assisted career guidance and professional development. Although the current implementation successfully achieves the project objectives, several enhancements can be incorporated in future versions to improve functionality, performance, scalability, and user experience. The following future enhancements are proposed.

1. Native Mobile Application

A dedicated mobile application for Android and iOS platforms can be developed using React Native or Flutter. The mobile application will provide an improved user experience for resume uploads, career roadmap tracking, mock interviews, AI chatbot interactions, and job recommendations. Push notifications can also be implemented to inform users about important updates and AI analysis results.

2. Multi-Language Support

The platform can be extended to support multiple languages for both the user interface and AI-generated content. This enhancement will allow users to access resume analysis, career roadmaps, interview preparation, and chatbot assistance in their preferred language, making the platform accessible to a wider range of users.

3. Advanced Recommendation System

The recommendation module can be enhanced by incorporating machine learning algorithms such as collaborative filtering and content-based recommendation techniques. This will enable the platform to provide more personalised job and course recommendations based on user behaviour, learning history, and career interests.

4. Video-Based Mock Interviews

The Mock Interview Module can be extended to support video-based interviews. Artificial Intelligence can evaluate facial expressions, eye contact, communication skills, speaking pace, and confidence levels in addition to analysing the content of user responses, providing more comprehensive interview feedback.

5. Email and SMS Notification System

The notification system can be expanded by integrating email and SMS services. Users will receive important updates such as resume analysis completion, administrator notifications, password reset requests, interview reminders, and career progress summaries through email or SMS.

6. Cloud Deployment and Scalability

The platform can be deployed on cloud infrastructure such as Amazon Web Services (AWS), Google Cloud Platform (GCP), or Microsoft Azure. Containerisation using Docker and orchestration with Kubernetes can improve scalability, availability, and system performance for large numbers of users.

7. Third-Party API Services

The platform can provide secure REST APIs for educational institutions, recruitment agencies, career guidance centres, and other organisations. These APIs will allow external applications to utilise resume analysis, career roadmap generation, interview evaluation, and recommendation services.

8. Progressive Web Application (PWA)

The web application can be enhanced as a Progressive Web Application (PWA) to support limited offline access. Users will be able to access previously viewed information, and data entered while offline can be synchronised automatically when an internet connection becomes available.

9. Advanced Role-Based Access Control (RBAC)

The existing authentication system can be enhanced by introducing additional user roles such as Super Administrator, Career Coach, Support Executive, Content Moderator, and Organisation Manager. Each role can be assigned specific permissions to improve administration, security, and management of institutional users.

10. Integration with Multiple Job Portals

The job recommendation module can be enhanced by integrating with multiple job portals such as LinkedIn, Naukri, Indeed, Glassdoor, and other recruitment platforms through their official APIs or authorised integration services. This enhancement will enable the platform to collect job opportunities from multiple sources, providing users with a wider range of relevant and up-to-date job recommendations. Integration with multiple job platforms will improve the accuracy, coverage, and reliability of the job recommendation module.

11. Integration with Multiple Online Learning Platforms

The learning recommendation module can be expanded by integrating with popular online learning platforms such as Coursera, Udemy, edX, LinkedIn Learning, Skillshare, and other educational providers through their official APIs or authorised services. This enhancement will allow the platform to recommend courses, certifications, and learning resources from multiple providers instead of relying primarily on YouTube. Users will benefit from a broader selection of high-quality learning materials that better match their career goals and identified skill gaps.

12. Expansion to Multiple Career Domains

Future versions of the platform can be extended to support careers beyond the Software and IT sector. The system can be enhanced to provide resume analysis, career roadmaps, interview preparation, job recommendations, and learning resources for domains such as Mechanical Engineering, Civil Engineering, Electrical Engineering, Electronics, Healthcare, Finance, Marketing, Human Resources, Law, Commerce, Management, and other professional fields. Expanding domain-specific AI models and knowledge bases will make the platform suitable for a broader range of students and job seekers.

CONCLUSION

The Carvion AI Career Development Platform was successfully designed, developed, and implemented as an Artificial Intelligence-based solution for career development and professional guidance. The project achieved its primary objectives by providing resume analysis, career roadmap generation, mock interview preparation, job recommendations, learning recommendations, and AI-assisted career guidance through a single platform.

The system integrates React, Django REST Framework, MongoDB, SQLite, Google Gemini 2.5 Flash API, and the JSearch API to deliver a secure, efficient, and user-friendly application. Its modular architecture improves maintainability, scalability, and supports future enhancements.

The major modules, including Resume Management, Career Roadmap, Mock Interview, Job Recommendations, AI Chatbot, and Administrator Module, functioned successfully and worked together to provide a complete career development solution. The integration of Artificial Intelligence improved resume evaluation, career guidance, interview assessment, and personalised recommendations.

The platform was evaluated through Unit Testing, Integration Testing, System Testing, Security Testing, Performance Testing, and User Acceptance Testing. The results confirmed that all major modules operated correctly and satisfied the functional and non-functional requirements of the project.

Although the current implementation has limitations, such as dependency on external AI services, limited job and learning resource integration, support primarily for IT-related careers, and the absence of a native mobile application, these limitations provide opportunities for future enhancements.

In conclusion, the Carvion AI Career Development Platform successfully demonstrates the practical application of Artificial Intelligence in career development. The project provides a reliable and user-friendly solution for career planning and has strong potential for future expansion through additional features, wider domain support, and improved AI capabilities.

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